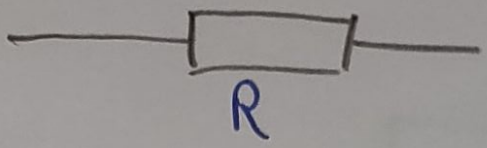
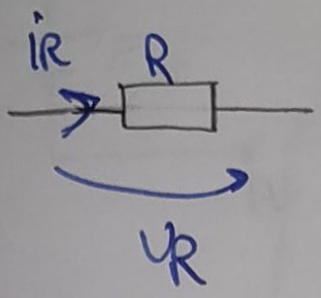
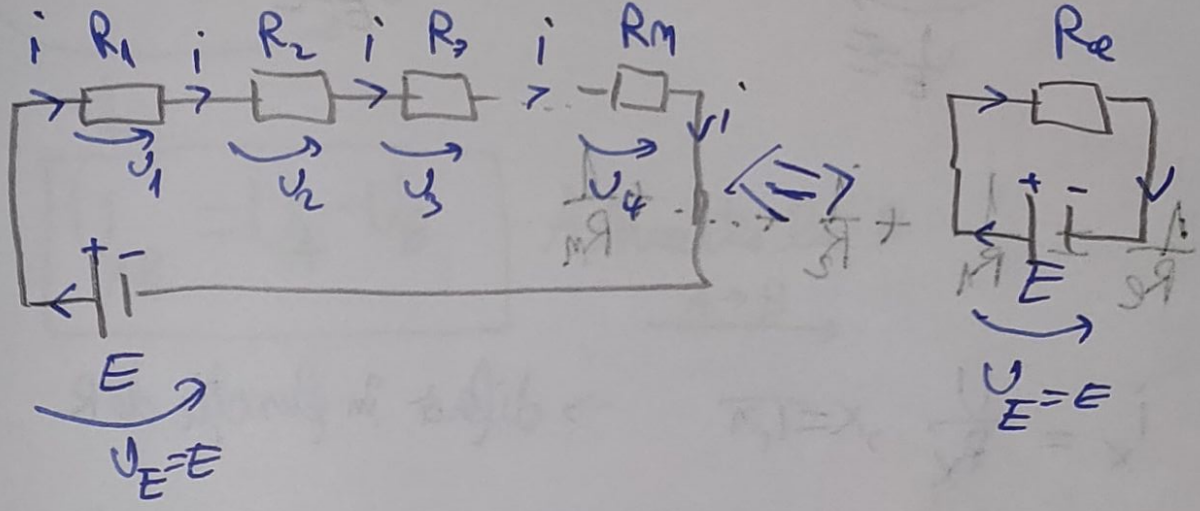
Rezistența





Legea lui Ohm: $\boxed{U = i \cdot R}$

- gruparea rezistențelor în serie

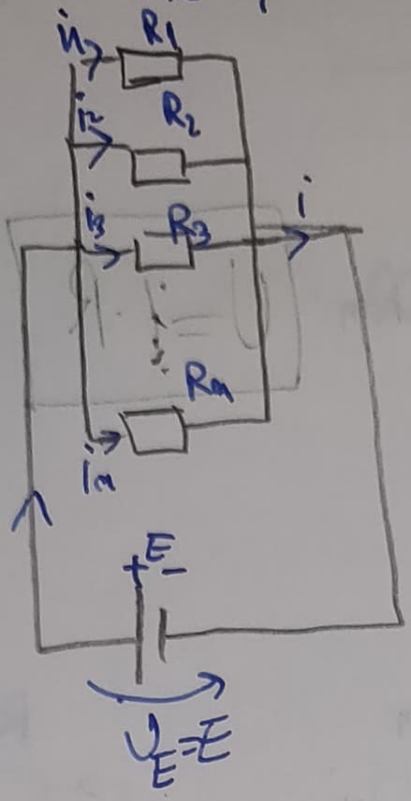


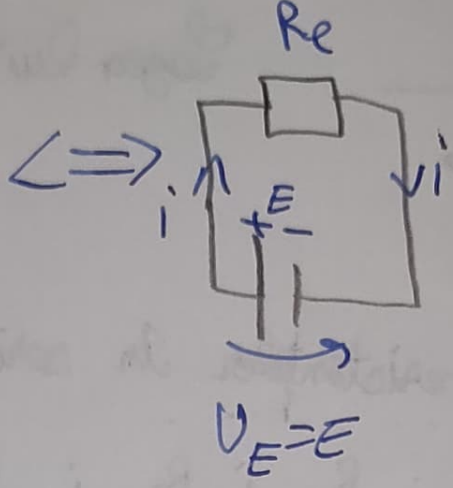
$$R_e = R_1 + R_2 + \ldots + R_n$$

$$i_{1\ldots n} = \frac{E}{R_e}$$ → este la fel în toate R

$$U_{1\ldots n} = i \cdot R_{1\ldots n}$$ → diferă în funcție de R

Gruparea rezistențelor în paralel





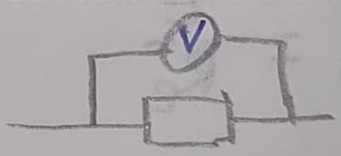
$$\frac{1}{R_e} = \frac{1}{R_1} + \frac{1}{R_2} + \ldots + \frac{1}{R_n}$$

$i_x = \frac{U}{R_x}$, $x = \overline{1,n}$ → diferă în funcție de R

$U_x = i \cdot R_e$, $x = \overline{1,n}$ → este la fel pt toate R-urile

---

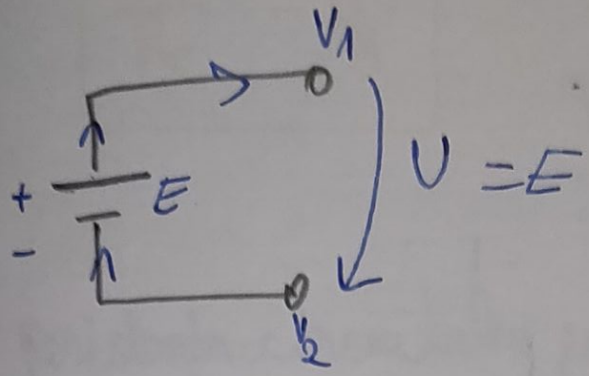
Voltmetrul
- măsoară tensiunea curentului
- se leagă în paralel



Ampermetrul
- măsoară intensitatea curentului
- se leagă în serie



Sursa de curent



$$U = V_1 - V_2$$



U_{AB}

$$U_{AB} = U_A - U_B$$

A înaintea lui B

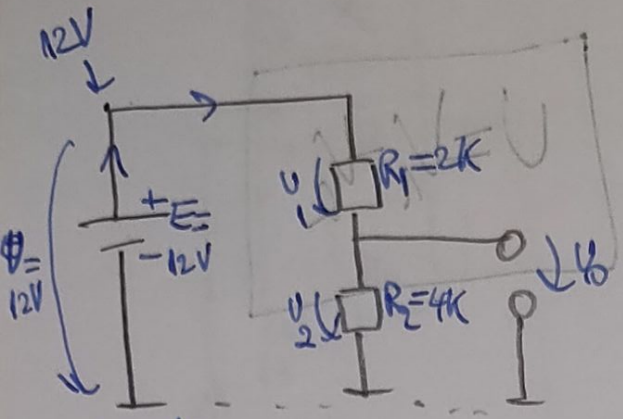
$A \rightarrow B$

Electronul e^- are sarcină negativă și circulă de la − la +

Intensitatea curentului electric I circulă de la + la −

• O diferență de potențial face electronul e^- să se miște de la potențialul mai mic, la cel mai mare

Divisor de tensiune



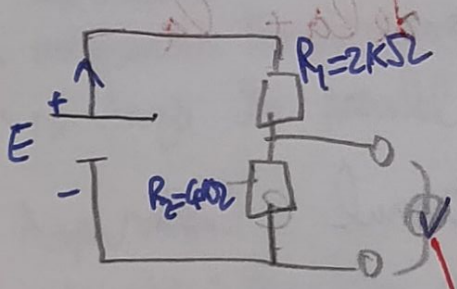
ground=
masă=
potențial 0=
0V

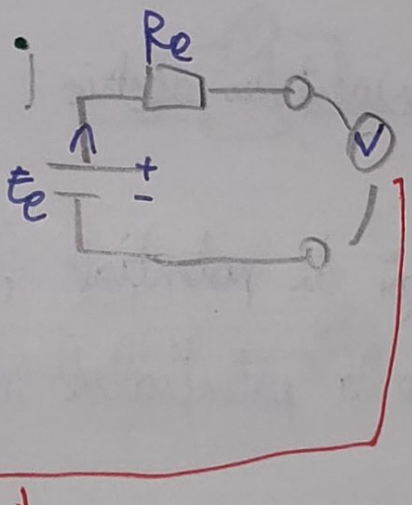
Il semnifică faptul că R_1 și R_2 sunt grupate în paralel

$R_e = R_1 \| R_2$

$$\frac{1}{R_e} = \frac{1}{R_1} + \frac{1}{R_2} \Rightarrow \frac{1}{R_e} = \frac{R_2 + R_1}{R_1 \cdot R_2} \Rightarrow R_e = \frac{R_1 \cdot R_2}{R_1 + R_2}$$

$$E_e = E \frac{R_2}{R_1 + R_2}$$

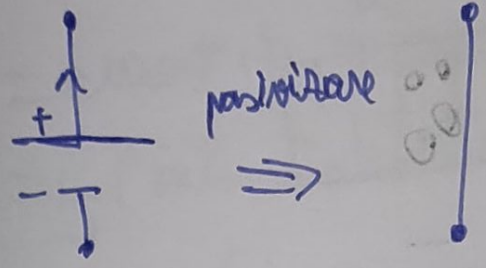


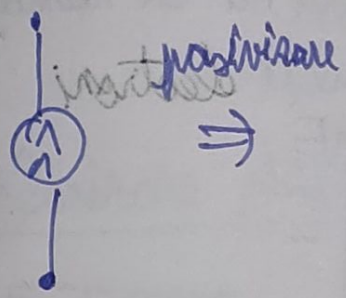


trebuie să indice aceeași tensiune

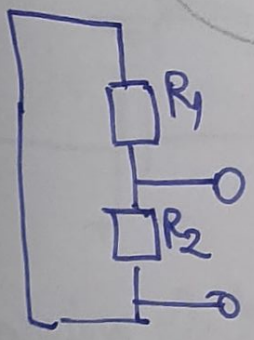
$$\frac{E_e}{R_e} = \frac{E}{R_1}$$

Rezistența circuitului pasivizat: $R_E = R_{PASIVIZAT}$





(izolator)

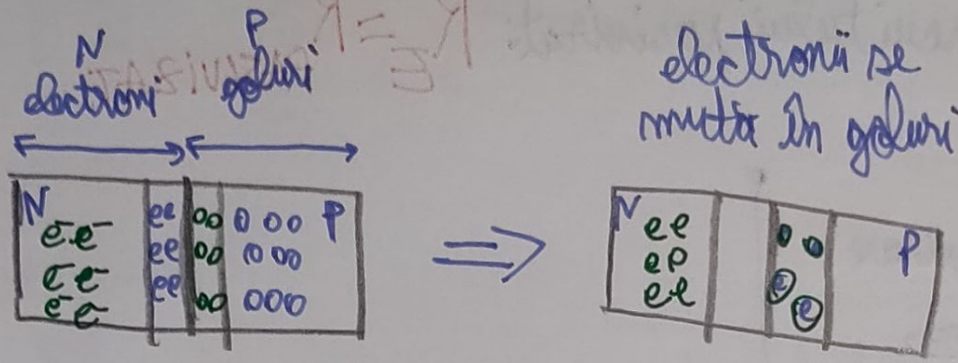


$$R_e = R_{PASIVIZAT} = R_1 \| R_2 = \frac{R_1 \cdot R_2}{R_1 + R_2}$$

Dioda → fluxul unidirecțional al curentului

Are două terminale

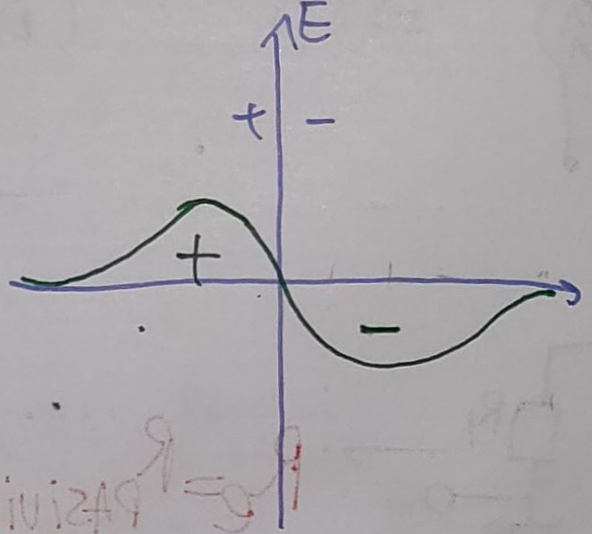
- anod (A) = semiconductor P (pozitiv) = goluri
- Katod (K) = semiconductor N (negativ) = e (electroni)

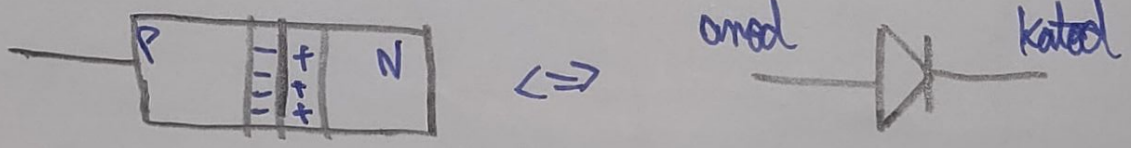


\+ —

pentru că au rămas mai mulți protoni

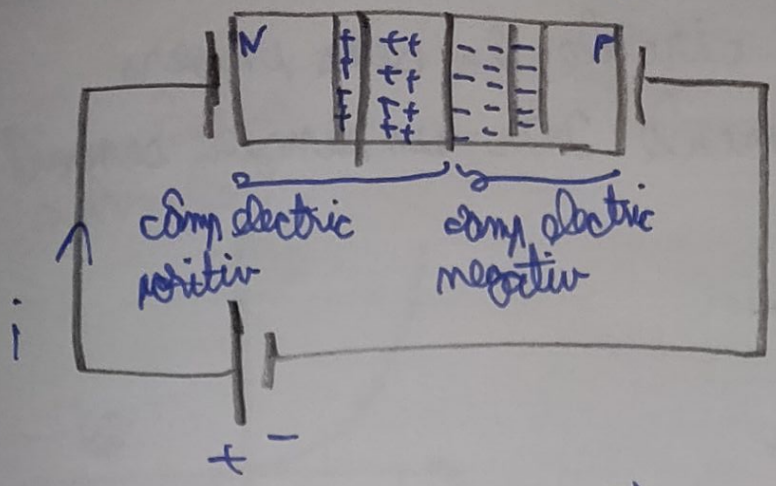
pentru că au rămas mai mulți electroni

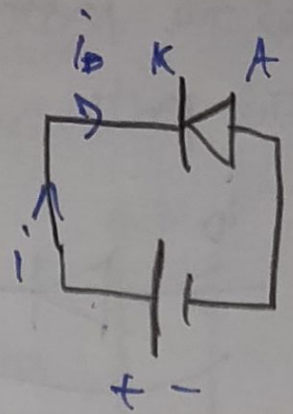




I. Polarizare inversă

Nu trece curent!





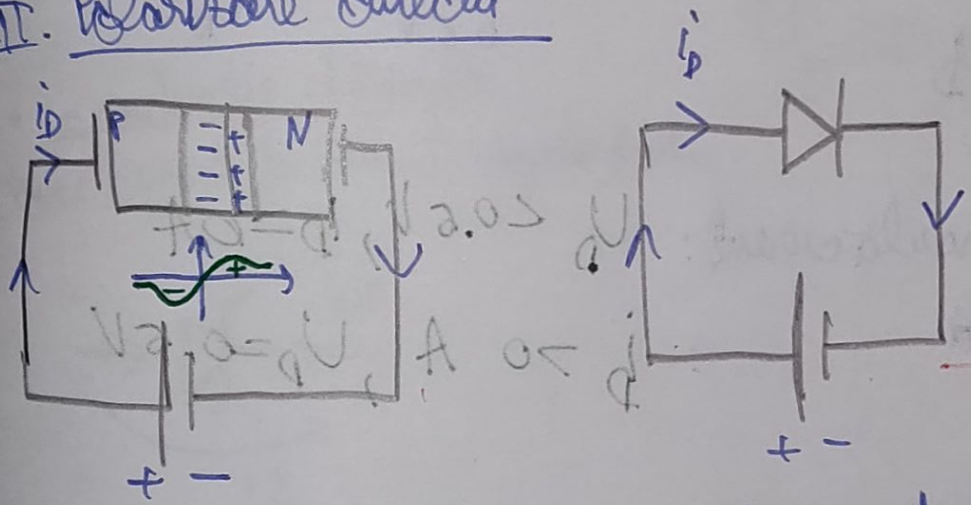
i = curent de generare (i_{gen})

$i = i_S$ (intensitatea de saturație)

$i_S \simeq$ pA (picoAmperi) / nA (nanoamperi)

10^{-12} 10^{-9}

II. Polarizare directă



Boltzman

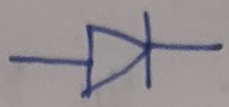
sarcina electrică: $\frac{q_0 \cdot U_D}{kT} \simeq \frac{1}{25} \cdot \frac{1}{mV}$

k = constanta lui ~~Boltzman~~

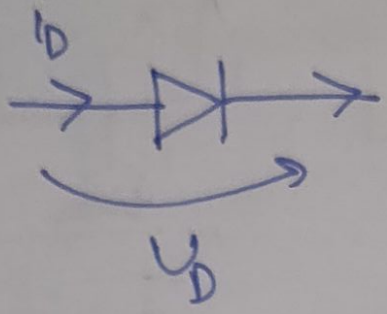
T = temperatura la care se află

$$i_D = i_S\left(e^{\frac{q_0 U_D}{kT}} - 1\right) \simeq i_S\, e^{\frac{1}{25\ mV} \cdot U_D}$$

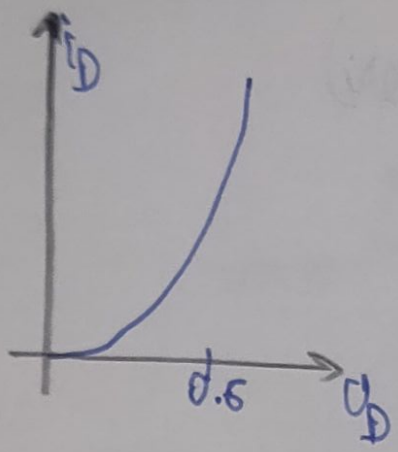
Dioda redresoare



- curentul nu poate să circule în sens invers
- curentul poate să circule într-un singur sens



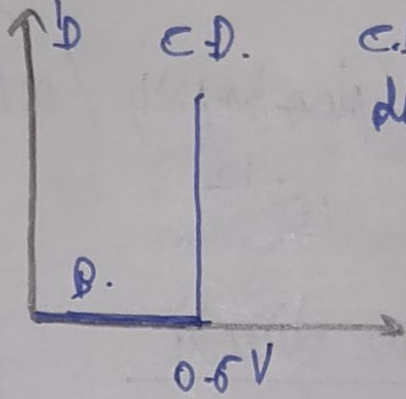
Dioda normală



Dioda ideală

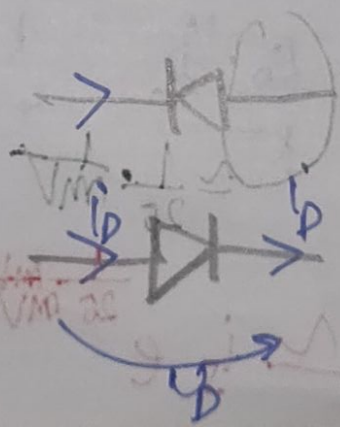
B = blocată

C.D. = conducție directă

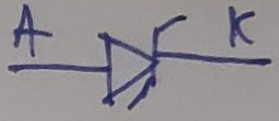


- blocată (nu circulă curent): $U_D < 0.6V$, $i_D = 0A$
- conducție directă (circulă curent): $i_D > 0$ A, $U_D = 0.6V$

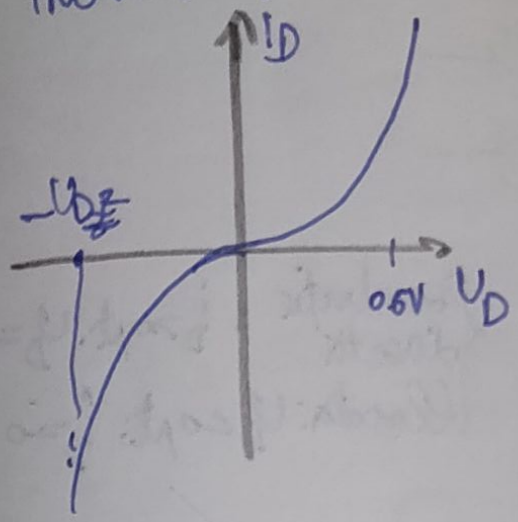
Blocată
conducție inversă



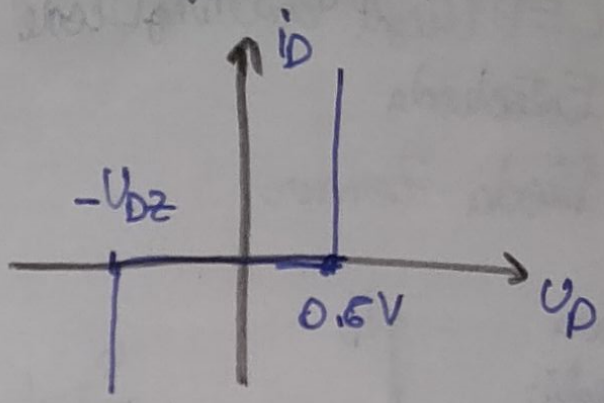
Dioda Zener



Dioda Zener normală



Dioda Zener ideală

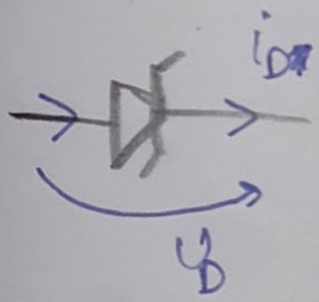


- <u>blocată</u> (nu circulă curent): $-U_{DZ} < U_D < 0.6V$, $i_D = 0$



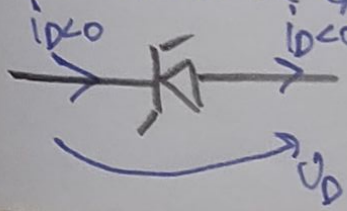
- <u>conducție directă</u> : $U_D = 0.6\,V$, $i_D > 0\,A$

(trece curent în sensul săgeții diodei)



- <u>conducție inversă</u> : $U_D = -U_{DZ}$, $i_D < 0$

(trece curent în sensul opus săgeții diodei)



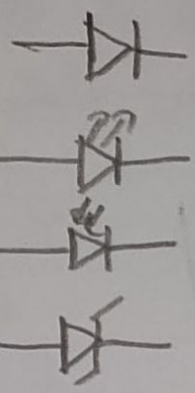
Tipuri de diode:

Dioda (redresoare)

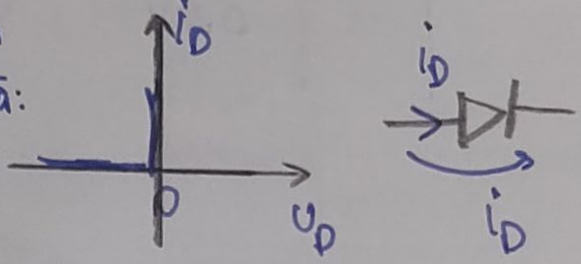
LED (Light Emitting Diode)

Fotodioda

Dioda Zenner



Dioda ideală:



Dioda ideală { conducție directă: $i_D > 0$ pt. $U_D = 0$
blocată: U_D oarecare pt. $i_D = 0$

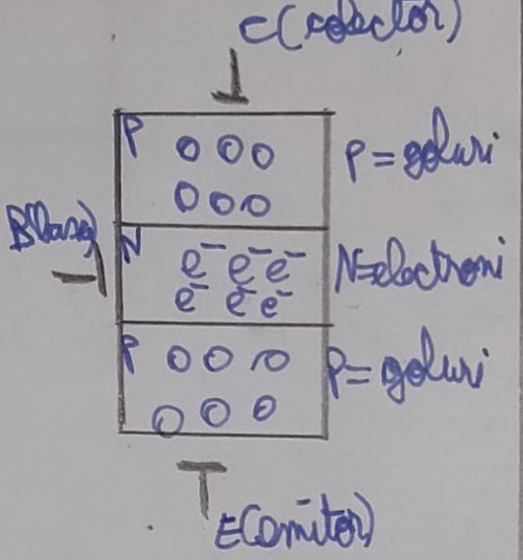
Tranzistorul → amplifică intensitatea curentului electric

B = baza (simbol: un "T" întors: "⊥")

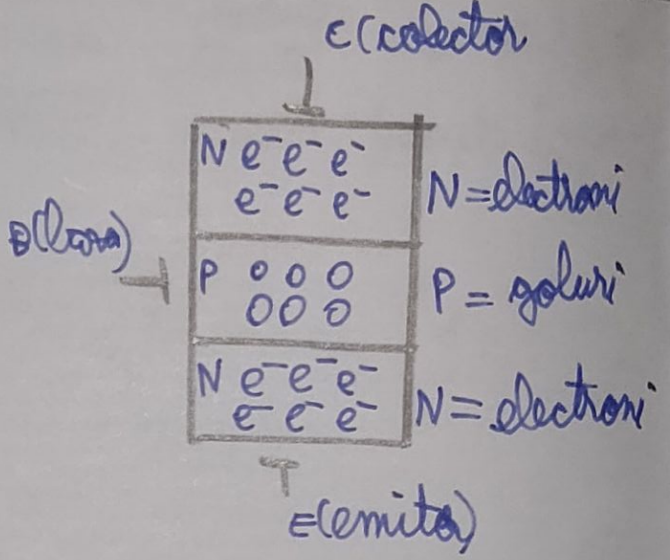
E = emitor (simbol: dreaptă cu o săgeată ↘ sau ↖)

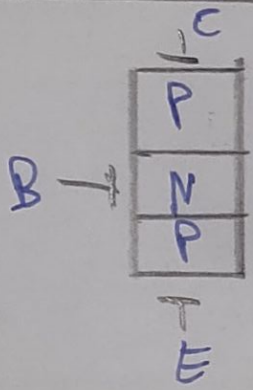
C = colector

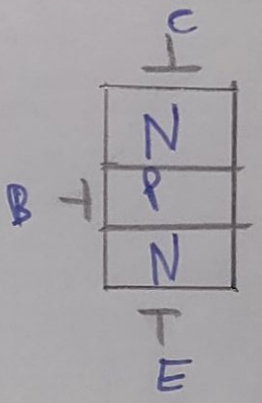
tranzistor PNP

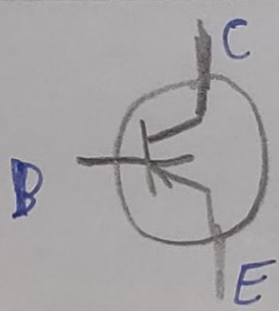


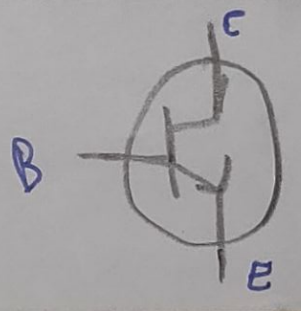
tranzistor NPN



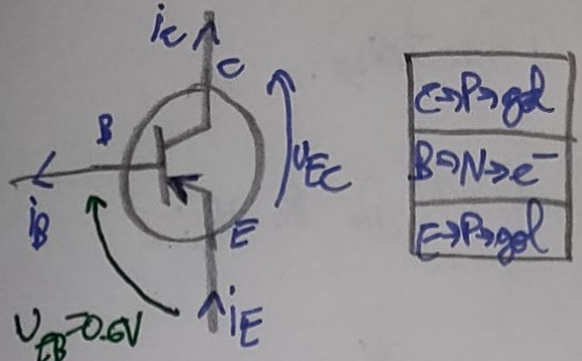








tranzistor PNP



i_E intră în tranzistor

$$i_E = i_C + i_B$$

RAN = Regiunea Activă Normală

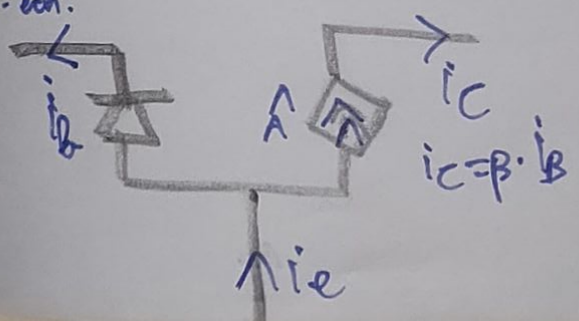
$$\text{RAN:} \begin{cases} U_{EB} = 0.6V \\ i_C = \beta\, i_B \\ U_{EC} \geq U_{EB} \end{cases}$$

$$\text{RAN} \Rightarrow i_E = i_C + i_B = i_C + \frac{i_C}{\beta} \simeq i_C$$

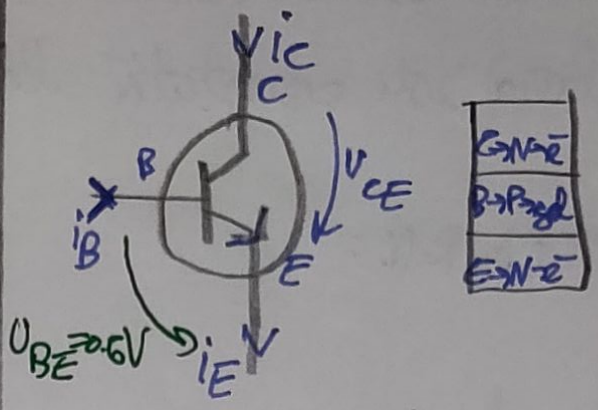
P.S.F. = Punctul Static de Funcționare

$i_C = ?$, $U_{EC} = ?$

circ. ech.



tranzistor NPN



i_E iese din tranzistor

$$i_E = i_C + i_B$$

RAN = Regiunea Activă Normală

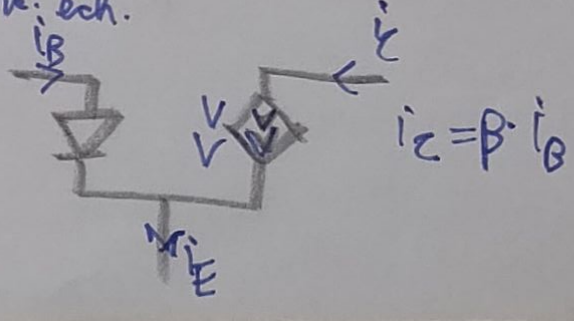
$$\text{RAN:} \begin{cases} U_{BE} = 0.6V \\ i_C = \beta \cdot i_B \\ U_{CE} \geq U_{BE} \end{cases}$$

$$\text{RAN} \Rightarrow i_E = i_C + i_B = i_C + \frac{i_C}{\beta} \simeq i_C$$

PSF = Punctul Static de Funcționare

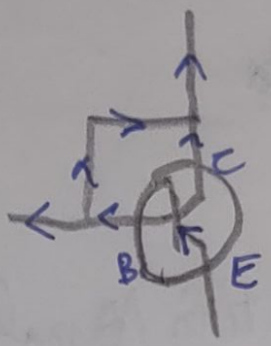
$i_C = ?$, $U_{CE} = ?$

circ. ech.

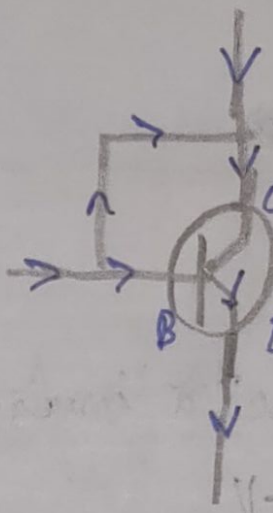


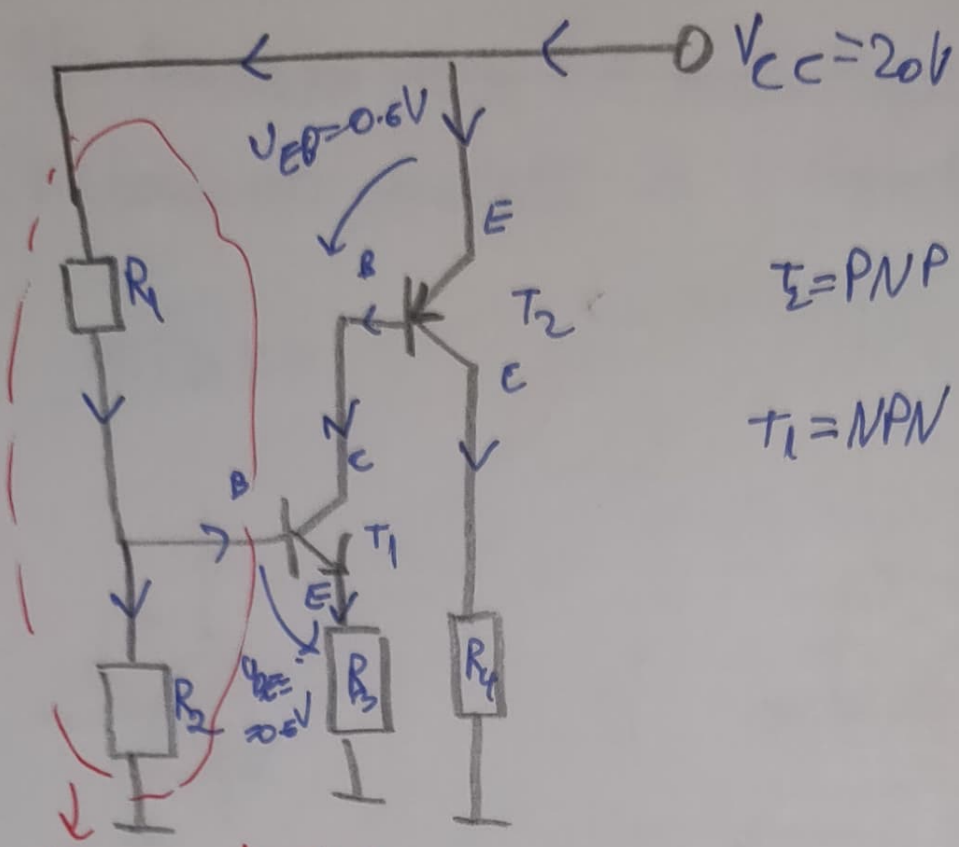
Un tranzistor este în R.N. (Reacție Negativă), dacă B (bază) este conectată la C (colector).

PNP în R.N.:



NPN în R.N.:





T_2 = PNP

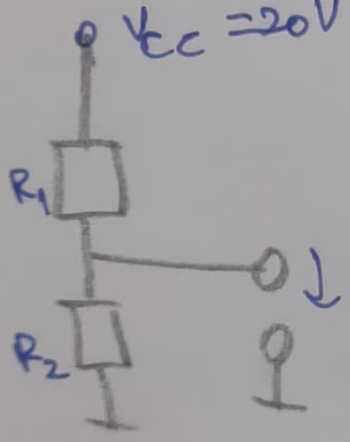
T_1 = NPN

divizor de tensiune

$V_{CC}=20V, R_1=5K\Omega, R_2=5K\Omega; R_3=4.4K\Omega, R_4=1.5K\Omega$

$\beta = 100$

PSF = ?



$$R_e = R_1 \parallel R_2$$

$$\frac{1}{R_e} = \frac{1}{R_1} + \frac{1}{R_2} \Rightarrow R_e = \frac{R_1 \cdot R_2}{R_1 + R_2}$$

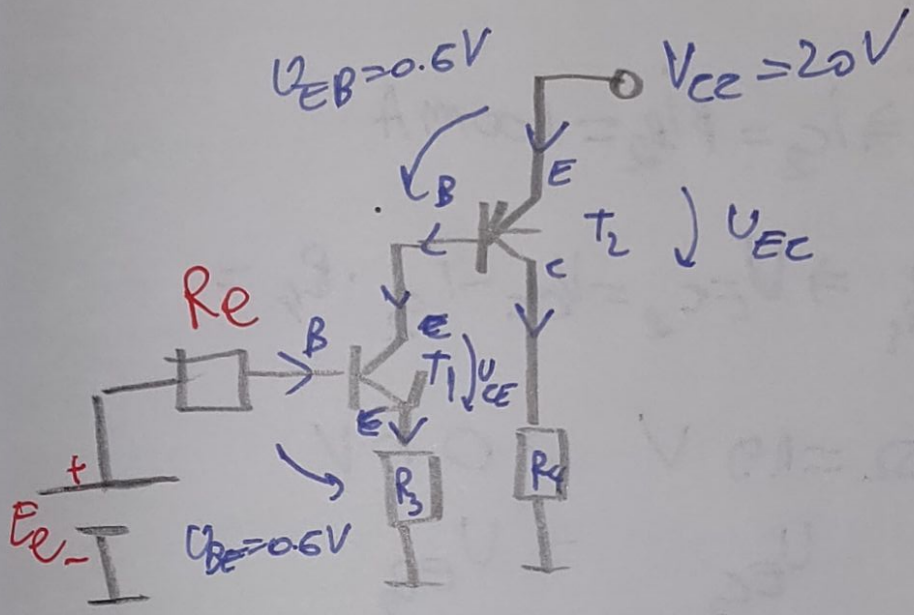
$$\frac{E_e}{R_e} = \frac{E}{R_1}$$

$$R_e = \frac{R_1 \cdot R_2}{R_1 + R_2} = \frac{15 \cdot 5}{15+5} = 3{,}75 K\Omega$$

$$\frac{E_e}{R_e} = \frac{E}{R_1} \Rightarrow \frac{E_e}{\frac{R_1 \cdot R_2}{R_1 + R_2}} = \frac{V_{cc}}{R_1} \Rightarrow E_e = \frac{R_1 \cdot R_2}{R_1 + R_2} \cdot \frac{V_{cc}}{R_1}$$

$$E_e = \frac{R_2}{R_1 + R_2} \cdot V_{cc} = \frac{10V}{15K\Omega + 5K\Omega} \cdot 5K\Omega = 5V$$

$R_e = 3{,}75 K\Omega$, $E_e = 5V$



Pentru a afla i_{C_1}, i_{C_2}, $U_{CE(T_1)}$, $U_{EC(T_2)}$ aplicăm legile lui Kirchhoff

$$E_e = i_{B_1} \cdot R_E + U_{BE} + i_{E_1} R_3$$

$$i_E = \frac{i_{C_1}}{\beta} \cdot R_E + U_{BE} + i_{C_1} \cdot R_3$$

$$5V = \frac{i_{C_1}}{100} \cdot 3{,}7K\Omega + 0.6V + i_1 \cdot 4.4K\Omega$$

$$\left(4.4 + \frac{3.75}{100}\right) i_{C_1} = 5V - 0.6V - 4.4V \Rightarrow$$

$$\Rightarrow i_{C_1} \simeq 1mA$$

$$i_{C_1} = i_{B_2} = 1mA \Rightarrow i_{C_2} = \beta i_{B_2} = 100mA$$

$$V_{CC} - 0 = U_{CE_2} + U_{R_1} \Rightarrow U_{EC_2} = V_{CC} - i_{C_2} \cdot R_4 =$$

$$= 20V - 100mA \cdot 10\Omega = 19V > 0.6V$$

$$U_{EC_2} > U_{EB_2}$$

$$V_{CC} - 0 = U_{EB_2} + U_{CE_1} + i_{e_1} R_3$$

$$20V = 0.6V + U_{CE_1} + 1mA \cdot 4{,}4K\Omega$$

$$20V = 5V + U_{CE_1}$$

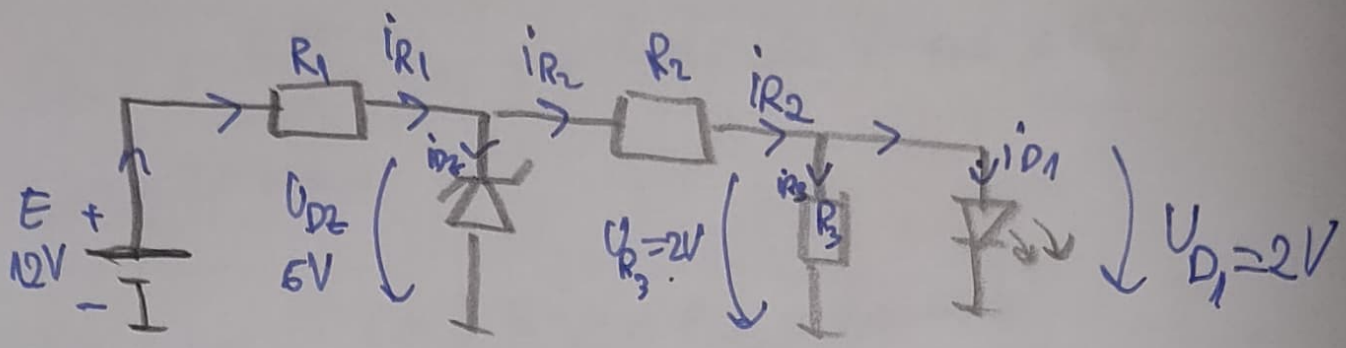
$$U_{CE_1} = 15V > 0.6V$$

$$U_{CE_1} > U_{BE_1}$$

1. $R_1 = 1K\Omega$, $R_2 = 1K\Omega$, $R_3 = 1K\Omega$

Se cunosc: $U_{DZ} = 6V$, $U_{D_1} = 2V$

$i_{D_1} = ?$ ($i_{LED} = ?$)



Kirchoff I:

~~$R_3 \| D_1 \Rightarrow$~~

~~$R_1 \| DZ \Rightarrow$~~

$$\begin{cases} R_3 \| D_1 \Rightarrow i_{R_2} = i_{R_3} + i_{D_1} \\ DZ \| R_2 \Rightarrow i_{R_1} = i_{DZ} + i_{R_2} \Rightarrow i_{R_1} = i_{DZ} + (i_{R_3} + i_{D_1}) \end{cases}$$

Kirchhof II: $E = U_{R_1} + U_{DZ} \Rightarrow E = i_{R_1} \cdot R_1 + U_{DZ} \Rightarrow$

$$\Rightarrow i_{R_1} = \frac{E - U_{DZ}}{R_1}$$

$$i_{R_1} = \frac{12V - 6V}{1K\Omega} = \frac{6V}{1K\Omega} = 6mA$$

Kirchhof II: $E = U_{R_1} + U_{R_2} + U_{R_3} \Rightarrow$

$$\Rightarrow E = i_{R_1} \cdot R_1 + i_{R_2} \cdot R_2 + i_{R_3} \cdot R_3$$

$R_3 \| D_1 \Rightarrow U_{R_3} = \underbrace{U_{D1}}_{2V} \Rightarrow U_{R_{R_3}} = 2V \Rightarrow i_{R_3} R_3 = 2V$

$$E = U_{R_1} + U_{DZ} \Rightarrow U_{R_1} = E - U_{DZ} = 12V - 6V = 6V$$

$$E = U_{R_1} + U_{R_2} + U_{R_3} \Rightarrow U_{R_2} = E - U_{R_1} - U_{R_3}$$

$$U_{R_2} = 12V - 6V - 2V \Rightarrow U_{R_2} = 4V \Rightarrow$$

$$\Rightarrow i_{R_{3-2}} \cdot R_{3-2} = 4V \Rightarrow i_{R_{3-2}} = \frac{4V}{R_{3-2}} \Rightarrow i_{R_{3-2}} = \frac{4V}{1K\Omega} = 4mA$$

Legea lui Ohm: $U_{R_3} = i_{R_3} \cdot R_3 \Rightarrow i_{R_3} = \frac{U_{R_3}}{R_3} = \frac{2V}{1K\Omega} = 2mA$

$i_{R_2} = i_{R_3} + i_{D_1} \Rightarrow 4mA = 2mA + i_{D_1} \Rightarrow \boxed{i_{D_1} = 1mA}$

Regim dinamic

Suprapoziție

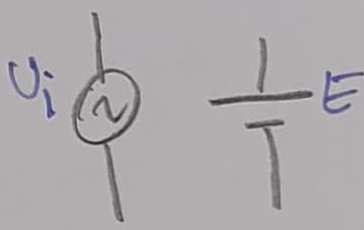
Dacă avem circuit liniar, cu componente liniare:

$$f(\alpha a + \beta b) = \alpha f(a) + \beta f(b)$$

$$f(u_1, u_2) = f(u_1, 0) + f(0, u_2)$$

↓ pasivizăm u_2 ↓ pasivizăm u_1

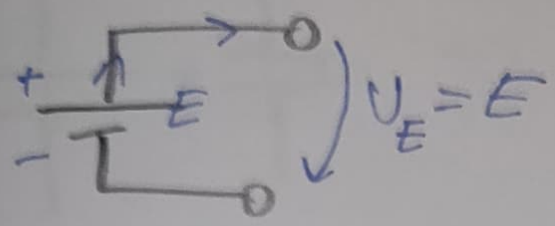
Tipuri de surse de intrare:



Când le avem pe amble (sau cel puțin două surse de intrare), nu putem calcula suma efectelor simultan. Astfel, se pasivizează pe rând.

Prin pasivizare, sursele pasivizate se vor transforma în fire.

• Sursa de tensiune (E) → generează curent continuu
Cântr-un singur sens



$$\frac{\Delta U_b}{\Delta t} \cdot 0 = 0$$

• Sursa de curent alternativ (U_i)
→ generează curent când cu +, când cu −
→ generează curent în ~~ambele~~ ambele sensuri

$$i(t) = A \cdot \sin(\omega \cdot t + \varphi_0)$$

$A = I_{max}$ = amplitudinea

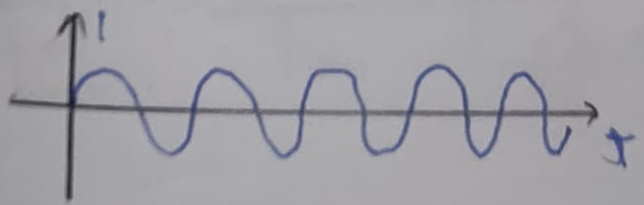
ω = pulsația, frecvența angulară

$\omega = 2 \cdot \pi \cdot f$

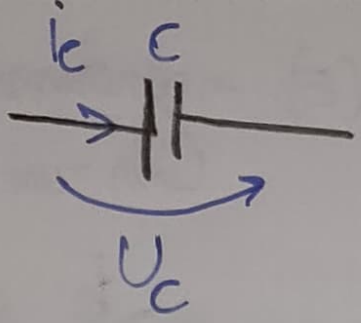
f = frecvența, $f = \frac{1}{T}$

la o frecvență $f = 50\ Hz$ ⇒ sensul curentului se inversează de $\frac{1}{f} = \frac{1}{50}$ de ori pe secundă

φ_0 = faza inițială, unghi $\in [0, 2\pi]$



Condensatorul



$$i_C = C \frac{dU_C}{dt}$$

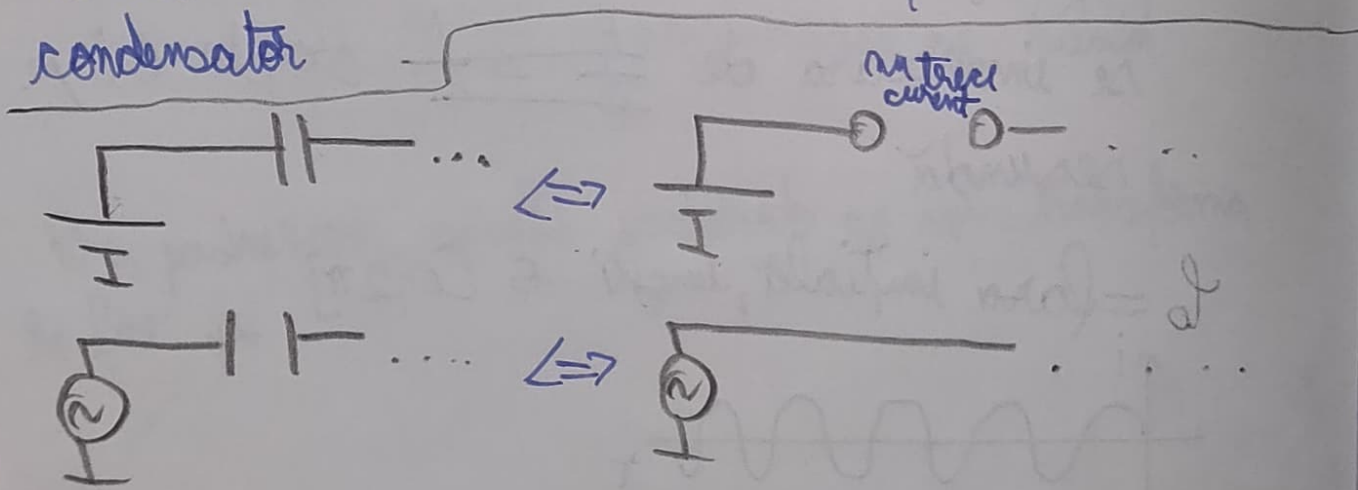
$i_C > 0 \Leftrightarrow U_i$ crește ↗

$i_C < 0 \Leftrightarrow U_i$ scade ↘

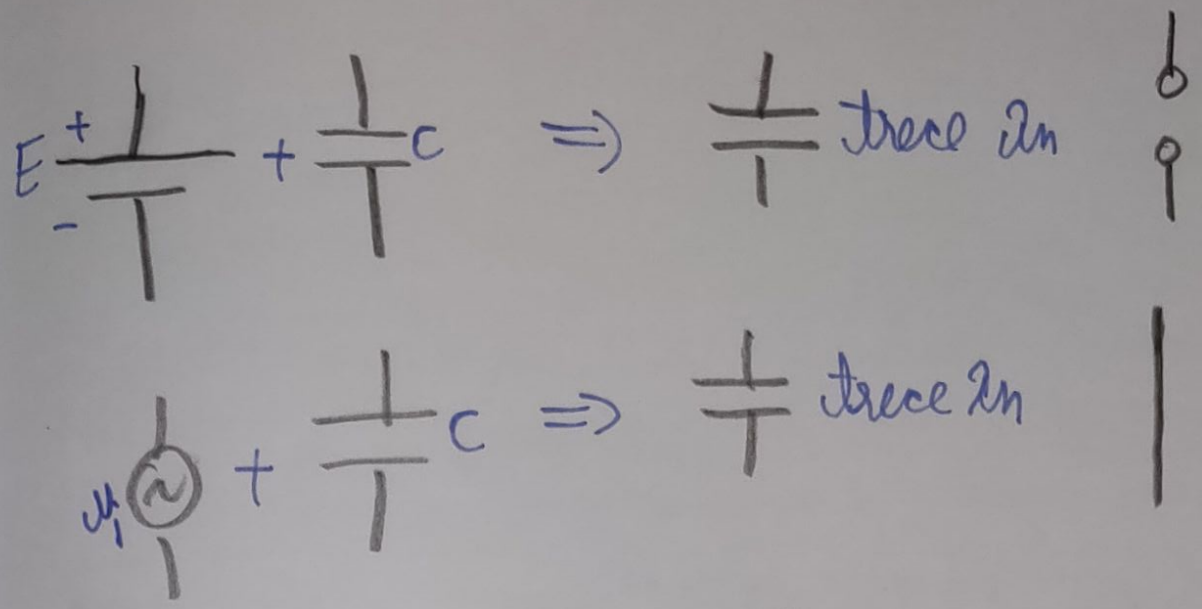
$i_C = 0 \Leftrightarrow U_i$ rămâne constant

Condensatoarele fac diferența între curentul continuu și curentul alternativ

- curentul continuu, generat de $\frac{+}{-}$ nu trece prin condensator
- curentul alternativ, generat de ~ trece prin condensator



Pasivizarea circuitului



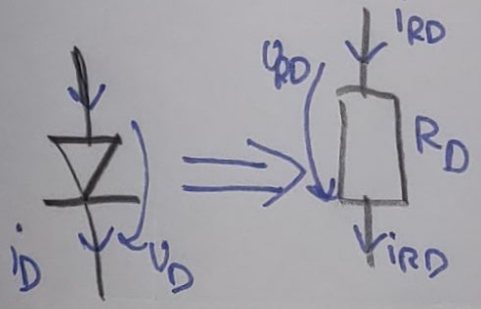
Cum liniarizăm o diodă?

curentul prin diodă

$$i_D = i_S\left(e^{\frac{q_0 U_D}{KT}} - 1\right) \approx i_S e^{\frac{q_0 U_D}{KT}}$$

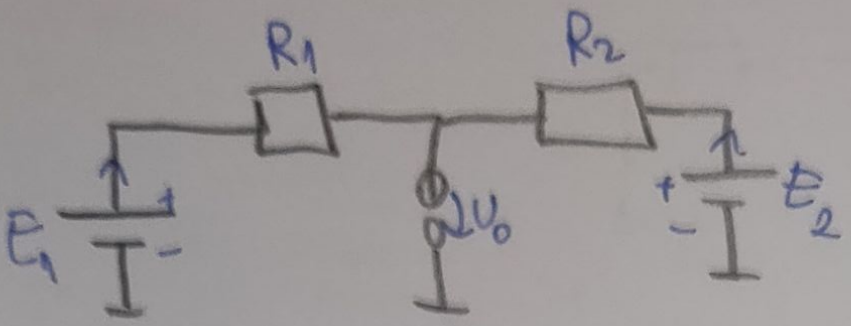
$$\frac{di_D}{dU_D} = \frac{di_S e^{\frac{q_0 U_D}{KT}}}{dU_D} = \frac{q_0}{KT}\left(i_D e^{\frac{q_0 U_D}{KT}}\right) = \frac{q_0}{K \cdot T} i_D$$

$$R_D = \frac{dU_D}{di_D} = \frac{KT}{q_0 \cdot i_D} \approx \frac{1}{40 \cdot i_D}$$



$$R_D = \frac{1}{40 \cdot i_D}$$

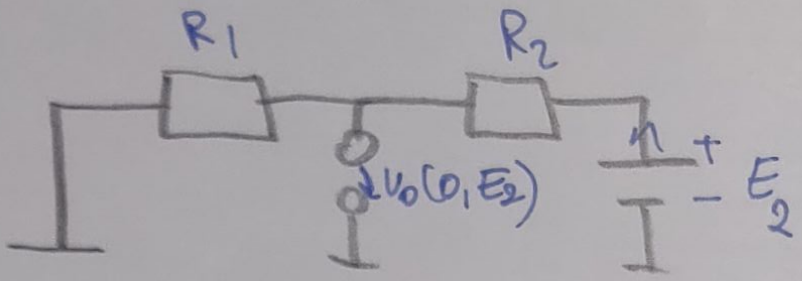
ex1 $U_0 = ?$



$U_0 = ?$

$U_0(E_1, E_2) = U_0(E_1, 0) + U_0(0, E_2)$ (liniarizare)

Pentru $U_0(0, E_2)$:



⇒ divizor de tensiune

~~$U_0(0, E_2) = \frac{E_e}{R_e} = \frac{E_2}{R_2}$~~ $U_0(0, E_2) = E_e \cdot \frac{E_e}{R_e} = \frac{E_2}{R_2}$

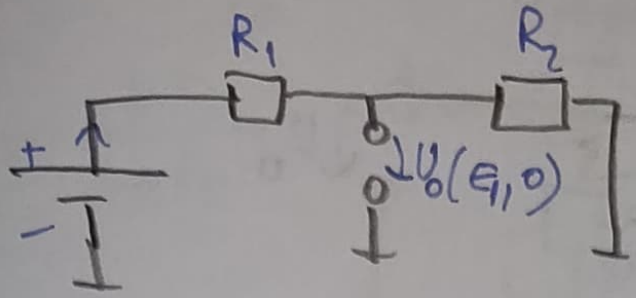
$R_e = R_1 \parallel R_2 \Rightarrow \frac{1}{R_e} = \frac{1}{R_1} + \frac{1}{R_2} \Rightarrow \frac{1}{R_e} = \frac{R_2 + R_1}{R_1 \cdot R_2} \Rightarrow$

$\Rightarrow R_e = \frac{R_1 \cdot R_2}{R_1 + R_2}$

$E_e = R_e \cdot \frac{E}{R_2} \Rightarrow E_e = \frac{R_1 \cdot R_2}{R_1 + R_2} \cdot \frac{E_2}{R_2} \Rightarrow E_e = \frac{R_1}{R_1 + R_2} \cdot E_2$

$U(0, E_2) = E_e = \frac{R_1}{R_1 + R_2} E_2$

Pentru $U_0(E_1, 0)$:



→ divizor de tensiune

~~$U_0(E_1, 0) = \frac{E_e}{R_e}$~~

$U_0(E_1, 0) = E_e$

$\frac{E_e}{R_e} = \frac{E_1}{R_1}$

$R_e = R_1 \| R_2 \Rightarrow \frac{1}{R_e} = \frac{1}{R_1} + \frac{1}{R_2} \Rightarrow \frac{1}{R_e} = \frac{R_1 + R_2}{R_1 \cdot R_2} \Rightarrow$

$\Rightarrow R_e = \frac{R_1 \cdot R_2}{R_1 + R_2}$

$E_e = R_e \cdot \frac{E_1}{R_1} \Rightarrow E_e = \frac{\cancel{R_1} \cdot R_2}{R_1 + R_2} \cdot \frac{E_1}{\cancel{R_1}} \Rightarrow$

$\Rightarrow E_e = \frac{R_2}{R_1 + R_2} \cdot E_1$

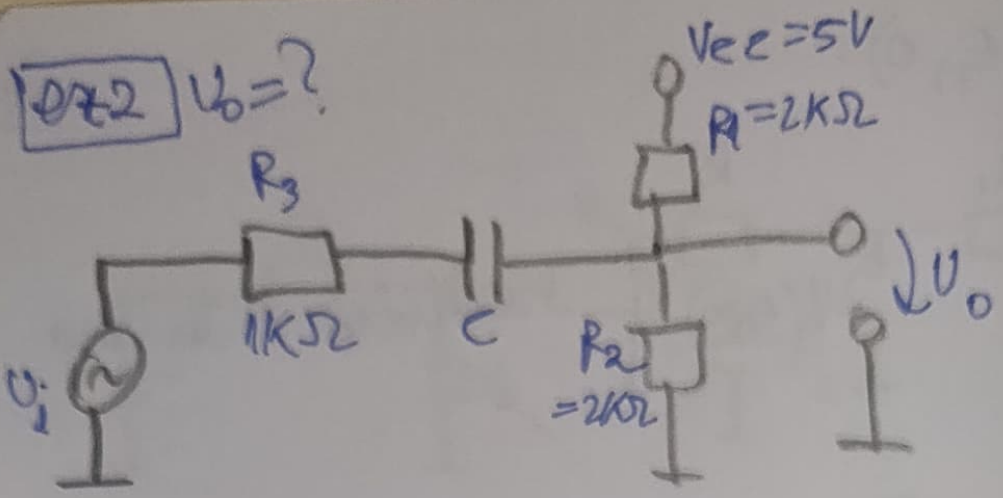
$U_0(E_1, 0) = E_e = \frac{R_2}{R_1 + R_2} E_1$

Finalizare

$U_0(E_1, E_2) = U_0(E_1, 0) + U_0(0, E_2) = \frac{R_2 E_1}{R_1 + R_2} + \frac{R_1 E_2}{R_1 + R_2} =$

$= \frac{R_1 E_2 + R_2 E_1}{R_1 + R_2}$

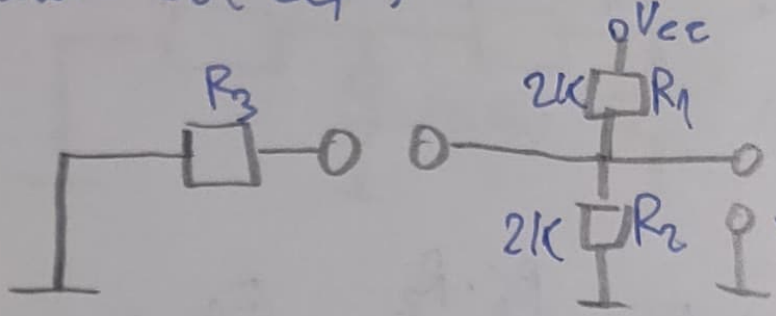
Ex 2 $U_0 = ?$



$U_0 = ?$

$U_0(V_{cc}, U_i) = U_0(V_{cc}, 0) + U_0(0, U_i)$ (liniarizat)

Pentru $U_0(V_{cc}, 0)$:



→ divizor de tensiune

$U_0(V_{cc}, 0) = E_e$

$\frac{E_e}{R_e} = \frac{V_{cc}}{R_2}$

$R_e = R_1 \| R_2 \Rightarrow \frac{1}{R_e} = \frac{1}{R_1} + \frac{1}{R_2} \Rightarrow \frac{1}{R_e} = \frac{R_1 + R_2}{R_1 \cdot R_2}$

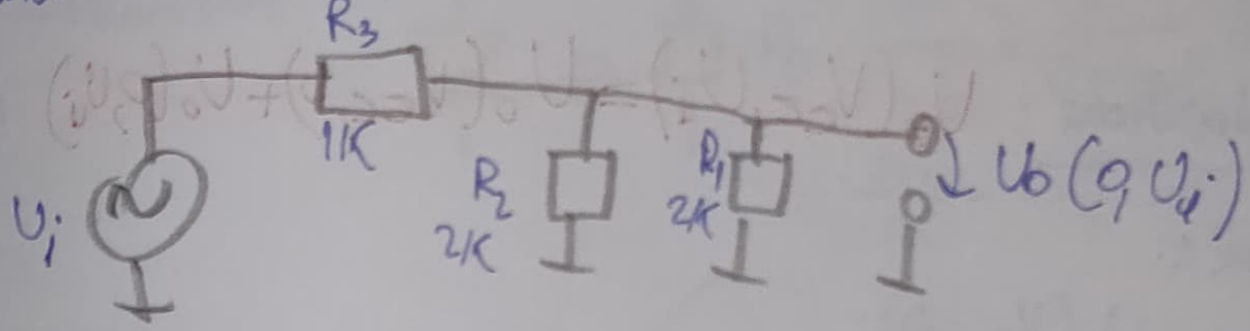
$R_e = \frac{R_1 \cdot R_2}{R_1 + R_2}$

$$E_e = R_e \cdot \frac{V_{cc}}{R_2} \Rightarrow E_e = \frac{R_1 \cdot R_2}{R_1 + R_2} \cdot \frac{V_{cc}}{R_2}$$

$$E_e = \frac{2K\Omega}{4K\Omega} \cdot 5V = 2{,}5V$$

$$U_0(V_{cc}, 0) = E_e \Rightarrow U_0(V_{cc}, 0) = 2{,}5V$$

Pentru $U_0(0, U_i)$:



$$U_0(0, U_i) = U_e$$

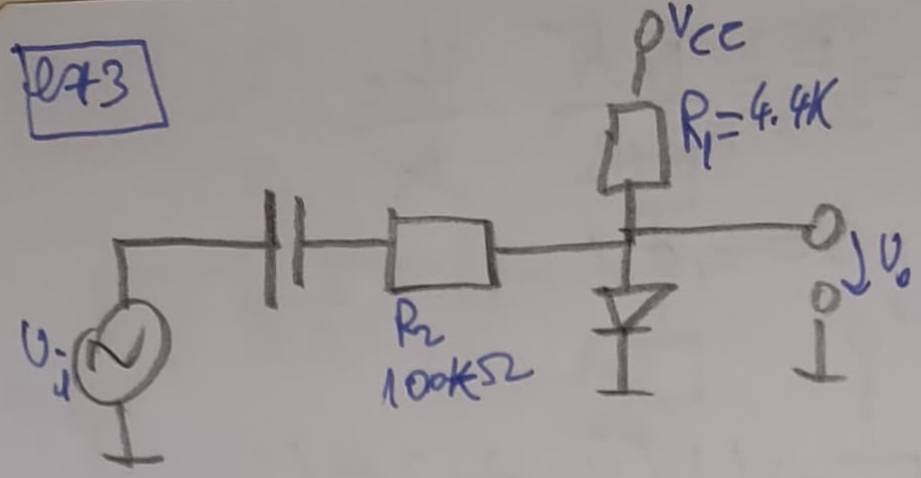
$$\frac{U_e}{R_e} = U_i$$

$$U_0(0, U_i) = \frac{R_1 \| R_2}{R_3 + R_1 \| R_2} U_i = \frac{1K}{1K + 1K} \cdot U_i = \frac{U_i}{2}$$

Finalizare

$$U_0(V_{cc}, U_i) = U_0(V_{cc}, 0) + U_0(0, U_i) = 2{,}5V + \frac{U_i}{2}$$

Ex 3

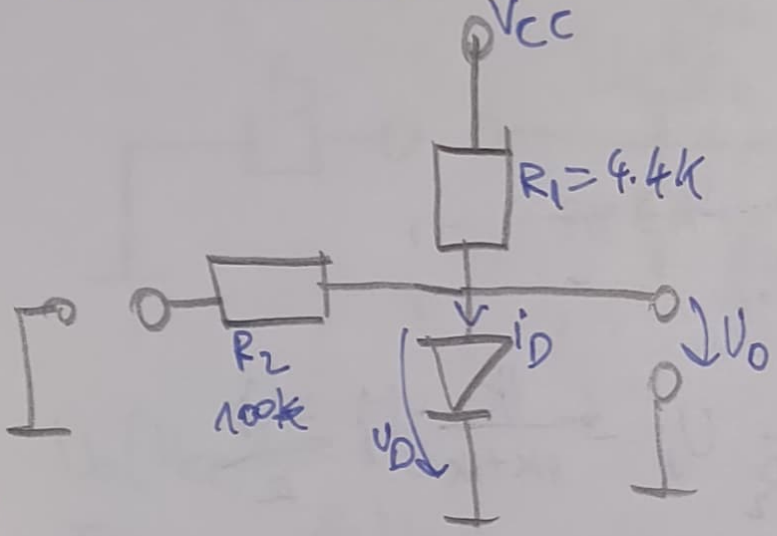


$U_0 = ?$

Liniarizare: $U_0(V_{CC}, U_i) = U_0(V_{CC}, 0) + U_0(0, U_i)$

Pasul 1:

Pentru $U_0(V_{CC}, 0)$:



- Pentru diodă: aprox. $U_D = 0.6V$

Presupunem D - conducție

U_D și U_0 sunt grupate în paralel $\Rightarrow U_D = U_0 \Rightarrow$

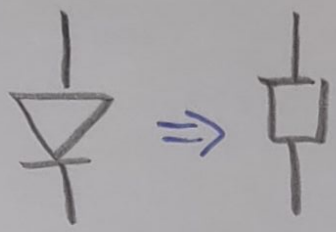
$\Rightarrow U_0 = 0.6V$

$V_{CC} = i_D \cdot U_D + R_1 \cdot i_D \Rightarrow I_D = \frac{V_{CC} - U_D}{R_1} = 1mA > 0$

$\Rightarrow$ dioda chiar este în conducție

Pasul 2:

liniarizare

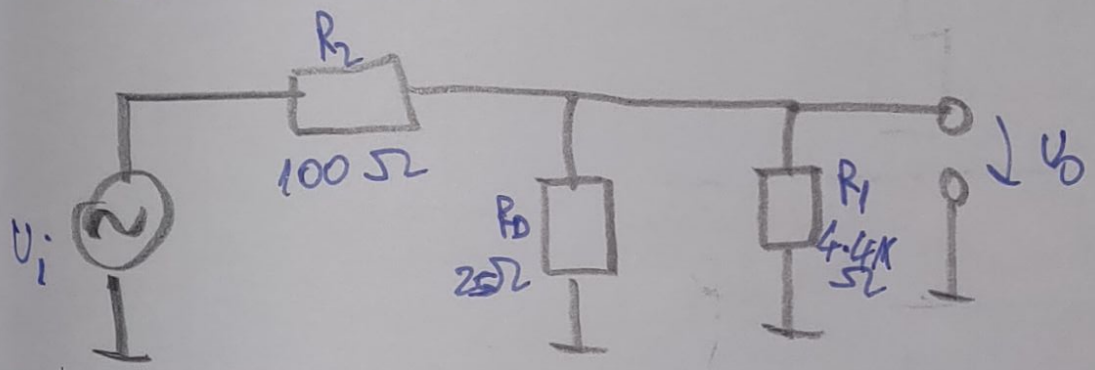


$R_D = \frac{1}{40 \cdot i_D} = \frac{1}{40 \cdot 1mA} =$

$= \frac{1000}{40} = \frac{100}{4} = 25\Omega$

Pasul 3:

Pentru $U_0(0, U_i)$



$R_1 \| R_2 = \frac{1}{\frac{1}{R_1} + \frac{1}{R_2}} = \frac{1}{\frac{R_1 + R_2}{R_1 \cdot R_2}} = \frac{R_1 \cdot R_2}{R_1 + R_2} =$

$= \frac{25 \cdot 4400}{25 + 4400} \simeq 25\Omega$

$$U_o(\text{q } U_i) = \frac{R_1 \| R_D}{R_1 \| R_D + R_2} \cdot U_i \cong \frac{25\Omega}{25\Omega + 100\Omega} \cdot U_i = \frac{U_i}{5}$$

Pasul 4 : finalizare

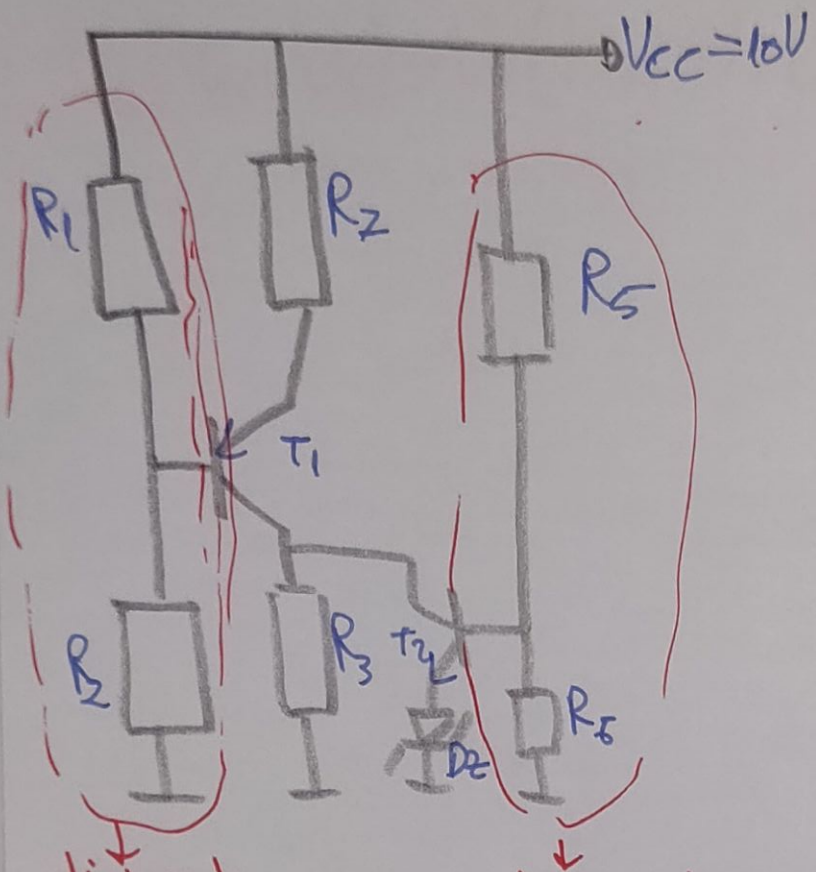
$$U_o(V_{CC}, U_i) = U_o(V_{CC}, 0) + U_o(0, U_i) = 0{,}6V + \frac{U_i}{5}$$

04

$V_{BE}=0.6V$, $\beta=100$, $V_D=0.6V$, $V_{DZ}=2.5V$, $V_{CC}=10V$

$R_1=2K\Omega$, $R_2=8K\Omega$, $R_3=5K\Omega$, $R_{4,5}=760K\Omega$

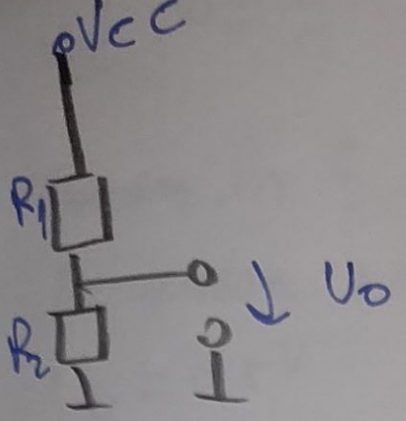
$R_6=760K\Omega$, $R_7=0.7K\Omega$



diviseur de tension (R_1 ; R_2)

diviseur de tension (R_5 ; R_6)

P.S.F. = ? (i_C = ? , V_{CE} = ?)



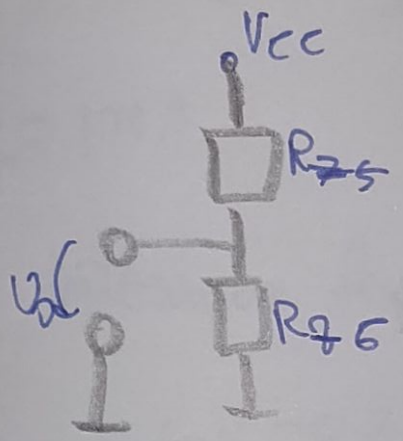
$$+\frac{1}{R_{e1}} = \frac{1}{R_1} + \frac{1}{R_2} \Rightarrow \frac{1}{R_e} = \frac{R_1 + R_2}{R_1 \cdot R_2}$$

$$R_{e1} = \frac{R_1 \cdot R_2}{R_1 + R_2} = \frac{1,6 \cdot 3}{2+3} = 1,6\,K\Omega$$

$$\boxed{\frac{V_{e1}}{R_{e1}} = \frac{V_{cc}}{R_1}}$$

$$V_{e1} = R_{e1} \cdot \frac{V_{cc}}{R_1} \Rightarrow V_{e1} = \frac{R_1 \cdot R_2}{R_1 + R_2} \cdot \frac{V_{cc}}{R_1}$$

$$V_{e1} = \frac{8}{2+3} \cdot 10 = \frac{8}{10} \cdot 10 = 8\,V$$



$$R_5 = R_6$$

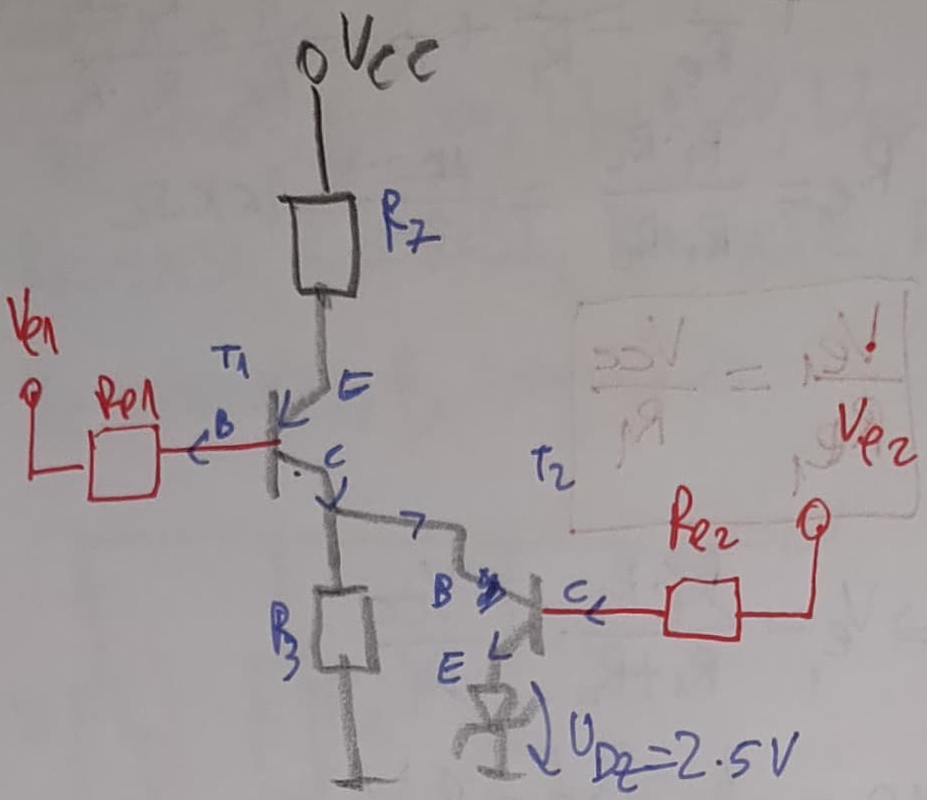
$$\frac{1}{R_{e2}} = \frac{1}{R_5} + \frac{1}{R_6} \Rightarrow \frac{1}{R_{e2}} = \frac{R_6 + R_5}{R_6 \cdot R_5}$$

$$R_{e2} = \frac{R_6 \cdot R_5}{R_6 + R_5} = \frac{R_5^2}{2 R_5} = \frac{R_5}{2}$$

$$\boxed{\frac{V_{e2}}{R_{e2}} = \frac{V_{cc}}{R_5}}$$

$$V_{e2} = R_{e2} \cdot \frac{V_{cc}}{R_5} \Rightarrow V_{e2} = \frac{R_5}{2} \cdot \frac{V_{cc}}{R_5} \Rightarrow V_{e2} = \frac{10}{2} = 5\,V$$

$$R_{e2} = \frac{R_5}{2} = \frac{760\,K\Omega}{2} = 380\,K\Omega$$



$$E_1 = i_{C_1} - i_{B_1} \simeq i_{C_1}$$

$$V_{CC} - V_{e_1} = R_z \cdot i_{C_1} + V_{EB_1} - R_{e_1} \cdot i_{B_1}$$

$$2V = 0{,}7k \cdot i_{C_1} + 0{,}6V - \frac{1.6k \cdot i_{C_1}}{100}$$

$$1.4V = i_{C_1}\left(0.7k - \frac{1.6}{100}k\right)$$

$$i_{C_1} \simeq \frac{1.4}{0.7k} = 2mA$$

$V_{CC} - 0 = R_2 \cdot i_{C_1} + U_{EC_1} + R_3 (i_{C_1} - i_{C_2})$

Trecem la T_2:

presupunem că dioda este polarizată direct

$U_D = 0.6\,V$

$V_{e_2} - 0\,V = R_{e_2} \cdot \frac{I_{C_2}}{100} + U_{BE_2} + U_D$

$5\,V = 330k \cdot \frac{I_{C_2}}{100} + 1.2\,V$

$3.8\,V = i_{C_2} \cdot \frac{330k}{100}$

$i_C = 1\,mA$

Ne interesăm la U_{EC_1}:

$U_{EC_1} = 10V - 0.2 \cdot 2 - 5 \cdot (2-1) = 3.6V$

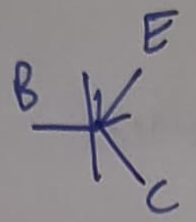
$V_{CC} - 0 = R_2 \cdot i_{C_1} + U_{EC_1} + i_{C_2} \cdot 0 + U_{CE_2} + U_D$

$10 = 1.4 + 3.6 + 0 + U_{CE_2} + 0.5$

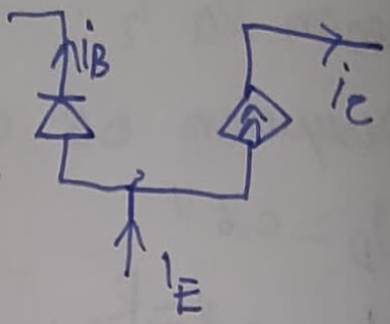
$U_{CE_2} = 4.4\,V$; $I_{C_2} = 1\,mA$

$U_{CEC_1} = 3.6\,V$; $i_{C_1} = ~~2m~~ 2\,mA$

→ Se transformă tranzistorul într-o rezistență și o sursă

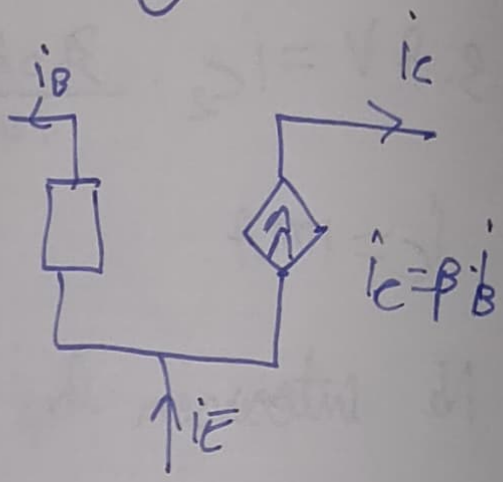


tranzistor PNP ⇒



$$R_D = \frac{1}{40 \cdot I_D}$$

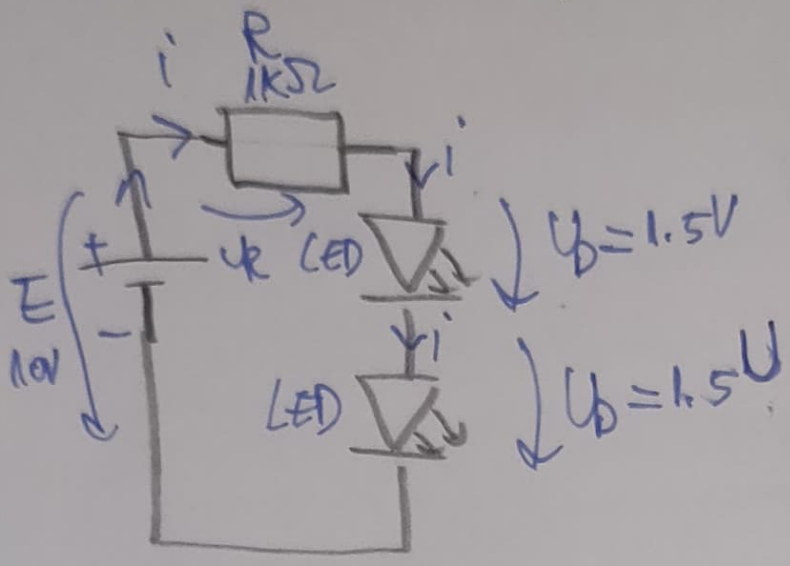
⇓



. Un tranzistor are R.N. (Reacție Negativă) dacă legăm Colectorul (C) la bază (B).

04

LED-urile din circuitul de mai jos sunt caracterizate de o cădere de tensiune $U_D = 1.5V$ și au un curent maxim permisibil de $i_D = 10mA$.



$i = ?$

$E = ?$ (intervalul de tensiune ca LED-urile să se aprindă fără să se ardă

$I = ?$

$E = U_R - U_D - U_D \Rightarrow U_R = E - 2 \cdot U_D \Rightarrow R \cdot i = E - 2 \cdot U_D \Rightarrow$

$\Rightarrow i = \frac{E - 2 \cdot U_D}{R} = \frac{10V - 2 \cdot 1,5V}{1k\Omega} = \frac{7V}{1k\Omega} = 7mA \leq$

$\leq 10mA$

pentru $i = 2mA$, LED-urile luminează fără să se ardă

$E = ?$

$E \in [E_{min}, E_{max}]$

Presupun că diodele vor avea $U_D = 1.5V$, indiferent de curent

$E \in [3V, E_{max}]$

$E_{max} = R \cdot 10mA + U_D + U_D \Rightarrow$

$E_{max} = 1K\Omega \cdot 10mA + 2 \cdot 1,5V$

$E_{max} = 13V$

$E \in [3V, 13V]$

? Se schimbă U_D în funcție de i_D ? NU pt. $i_D \in [0mA, 10mA]$

stări LED: { blocată: $i_D \leq 0mA \Rightarrow U_D = 0V$
conducție directă: dacă $0mA < i_D \leq 10mA$, at. $U_D = 1.5V$
dacă $i_D > 10mA$, at LED-urile se ard

$E = ?$. LED-urile sunt în conducție directă

$E = U_R + 2 \cdot U_D \Rightarrow E = i \cdot R + 2 \cdot 1,5V \Rightarrow E = i_D 1K\Omega + 3V$

$i_D > 0 \Rightarrow i_D \cdot 1K\Omega + 3V > 0 + 3V \Rightarrow E > 3V$

$i_D \leq 10mA \Rightarrow i_D \cdot 1K\Omega + 3V \leq 10mA \cdot 1K\Omega + 3V \Rightarrow E \leq 10V + 3V \Rightarrow E \leq 13V$

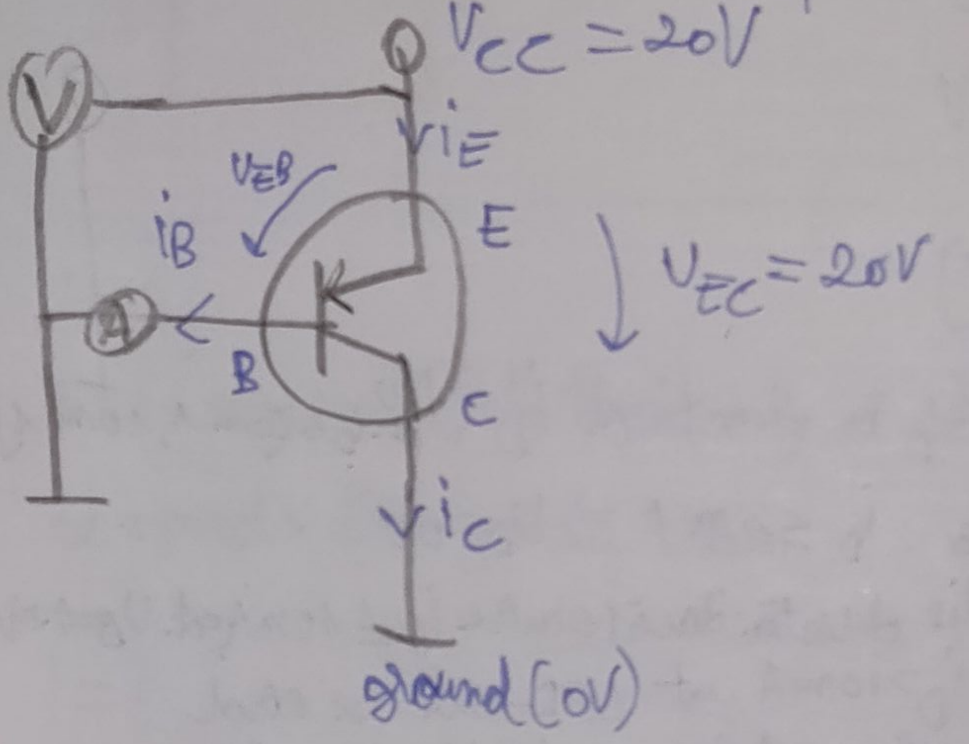
$\Rightarrow E \in (3V, 13V]$: $3V < E \leq 13V$

Q4

Desenați un circuit care să permită să măsurați caracteristica $I_B(U_{EB})$ a unui T. PNP pentru $U_{EC} = 20V$.

Explicați circuitul și metoda de măsură.

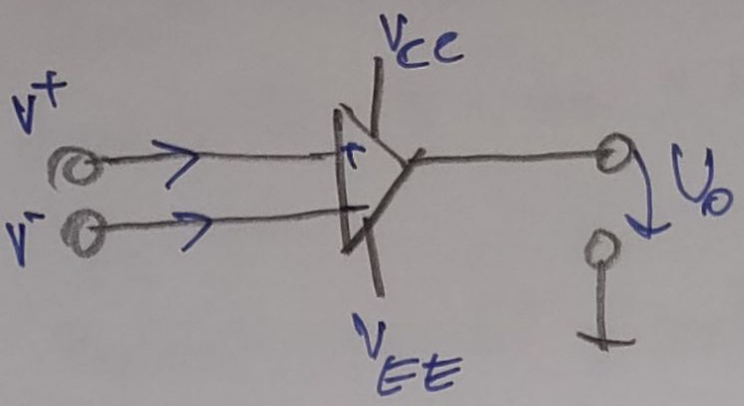
Desenați și explicați grafical ce care vă așteptați să îl obțineți



Pentru a măsura i_B, am legat un amperметru (A) în serie cu baza (B) tranzistorului.

Pentru a măsura U_{EB}, am legat un voltmetru (V) în paralel cu tranzistorul, dată înainte de emitor (E) și de bază (B).

Amplificatorul Operațional (A.O.)



V^+ = intrarea fără inversare

V^- = intrarea cu inversare

V_{CC} = tensiune Colector-Colector

V_{EE} = tensiune Emitor - Emitor

V_{PP} = peak-to-peak value (semnale oscilante, sin)

$V_{PP}(t) = A \cdot \sin(2\pi f t + \Phi)$, f = frecvență, t = timpul

U_0 = tensiunea de ieșire

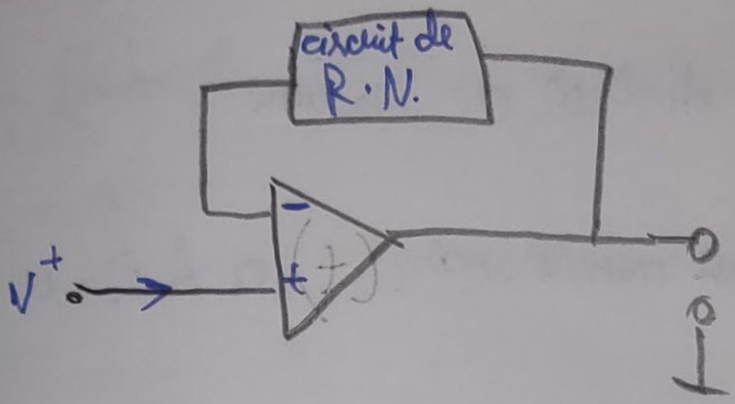
$$U_0 = A(V^+ - V^-)$$

Tipuri de A. O. - uri:

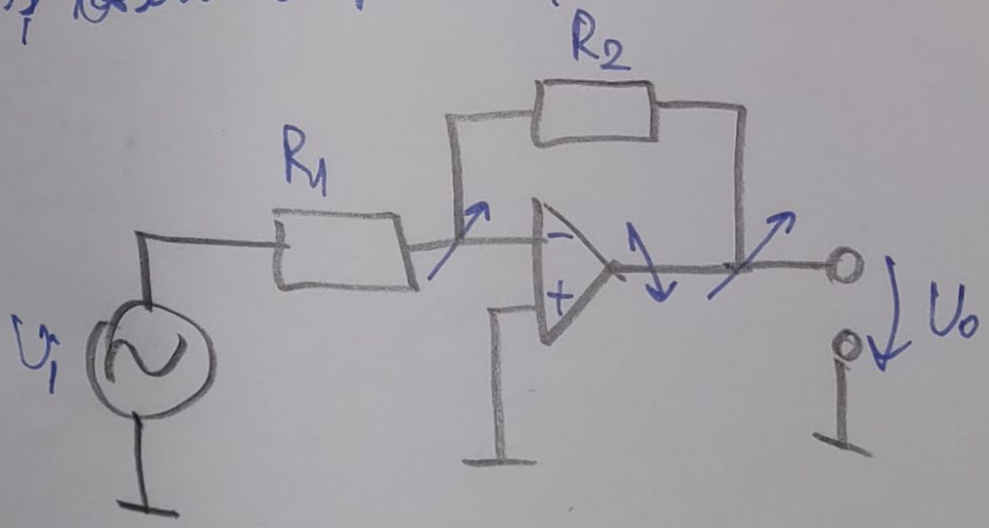
inversor, neinversor, repetor, sumator, log, exp

R.N. (Reacție Negativă) pentru A.O. $\Rightarrow V^+ = V^-$

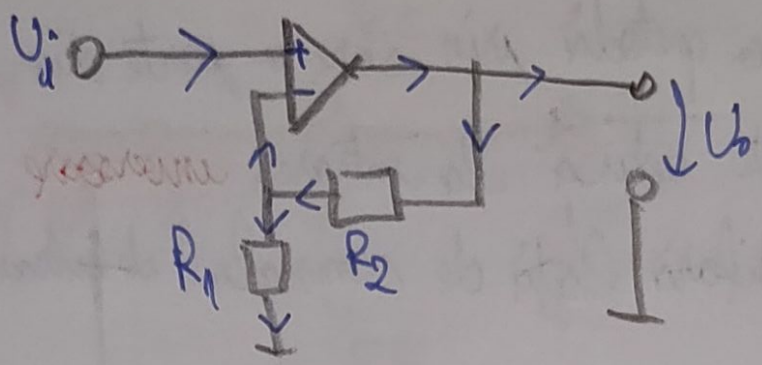
Reacția negativă este o metodă prin care o parte din tensiunea de ieșire U_o este adusă la intrarea inversoare a unui A.O., în antifază față de semnalul de intrare



- tensiunea de ieșire este direct proporțională cu tensiunea de intrare
- verificăm dacă reacția este negativă, crescând U_b și observând perturbațiile



A.N. (Amplificator Operational Normal)

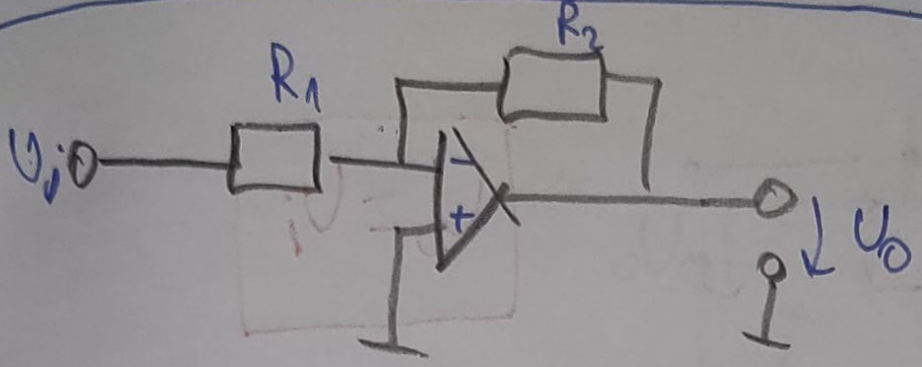


$$U_o = \frac{R_1 + R_2}{R_1} \cdot U_1$$

R_1 și R_2 formează un divizor de tensiune

. U_i polarizează borna ~~inv~~ neinversoare (+) a A.O.-ului

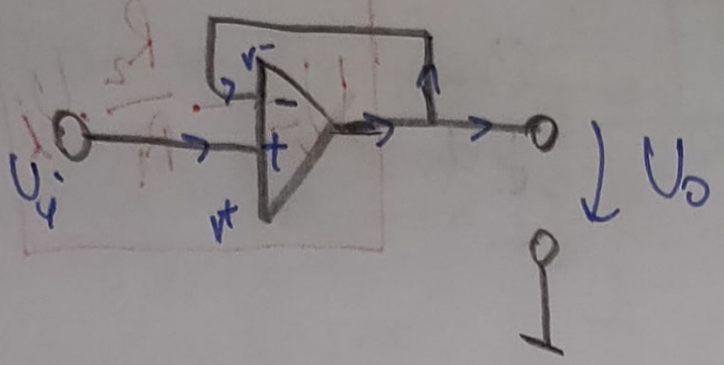
A.i. (Amplificator operational inversor)



$$U_o = -\frac{R_2}{R_1} \cdot U_i$$

. U_i folosește borna inversoare (−) a. A.O.-ului.

A.O. Repetor de tensiune



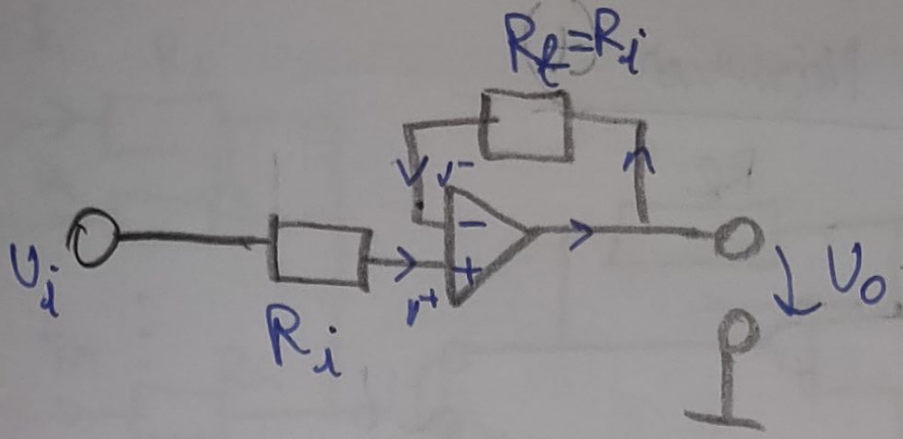
$$U_o = U_i$$

Conexiunea A.O. repetor constituie un caz particular de A.O. neinversor, în care întreaga tensiune de ieșire este adusă la intrarea inversoare prin conectare directă.

A.O. este în R.N. (Reacție Negativă) $\Rightarrow V^+ = V^-$ $\Rightarrow U_o = U_i$

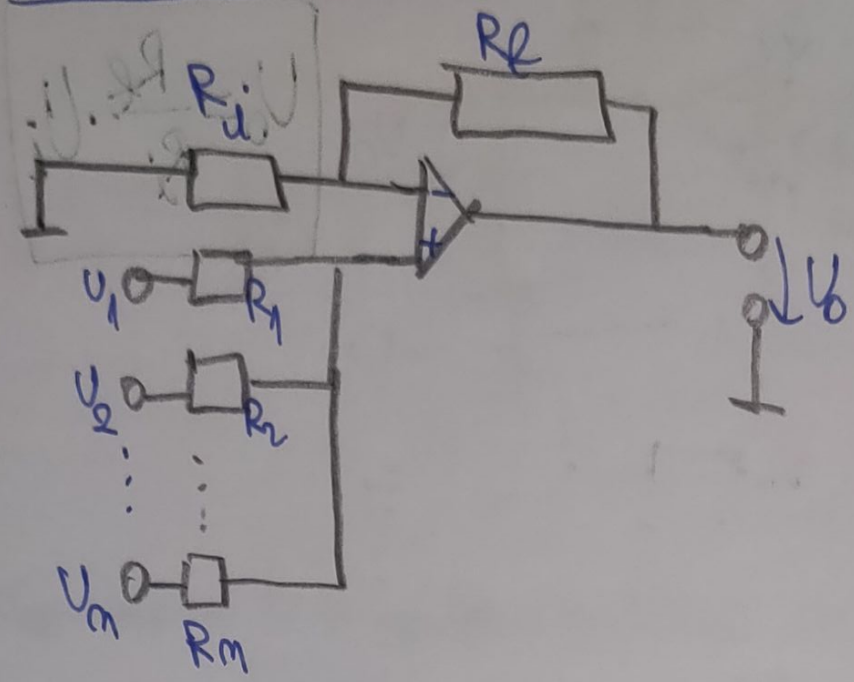
$V^+ = U_i$

$V^- = U_o$

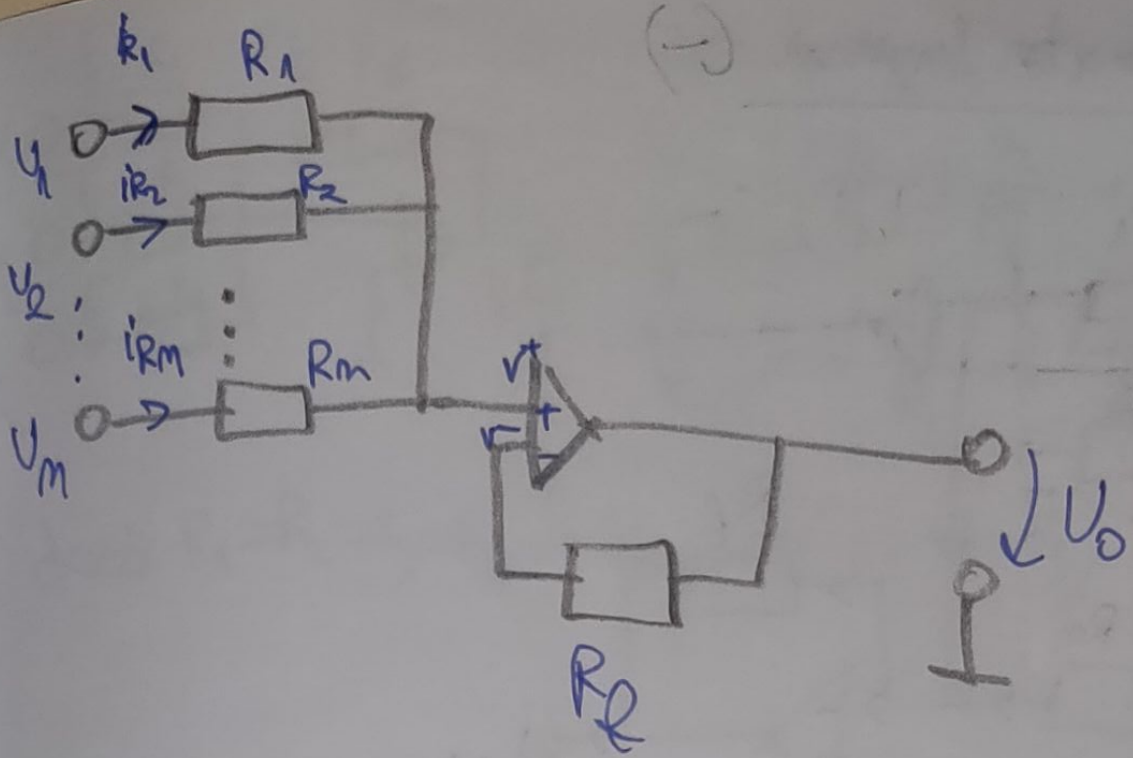


$$V_o = \frac{R_f}{R_i} \cdot V_i$$

A.O. Sumator Neinversor (+)



Dacă $\begin{cases} R_1 = R_2 = \dots = R_m = R \\ R_i = R_f = 2 \cdot R \end{cases} \Rightarrow U_0 = \sum_{i=1}^{m} U_i$



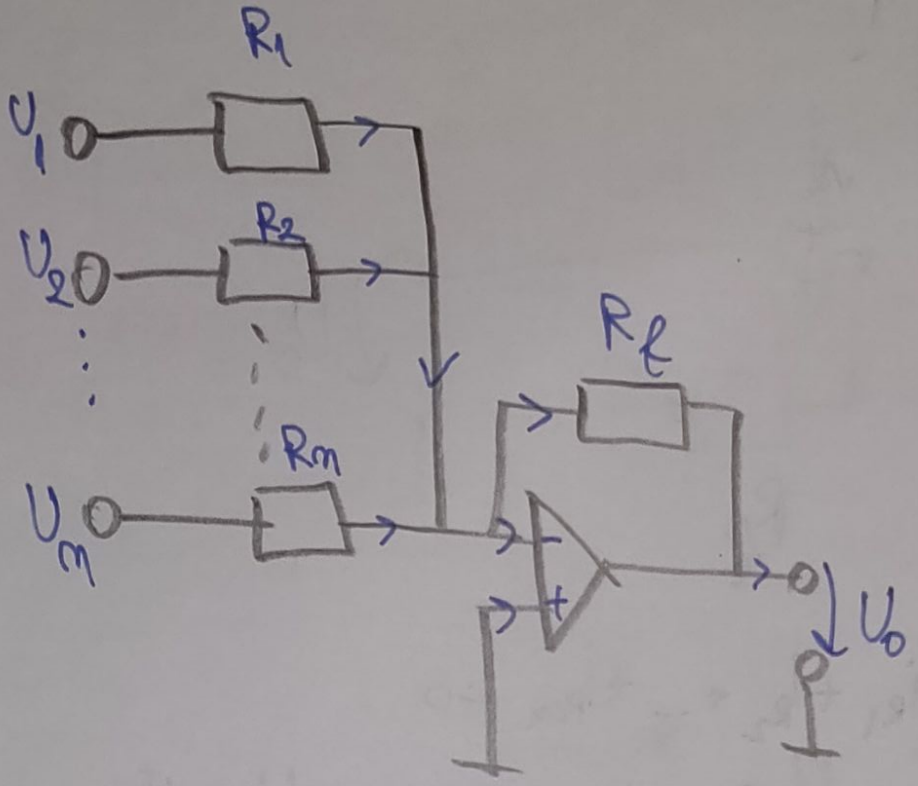
pentru intrarea cu + :

$$\begin{cases} i_{R_1} + i_{R_2} + \dots + i_{R_m} = 0 \\ \dfrac{U_1 - V^+}{R_1} + \dfrac{U_2 - V^+}{R_2} + \dots + \dfrac{U_m - V^+}{R_m} = 0 \end{cases}$$

$$U_{out} = \frac{R_f}{R_p}\left(U_1 + U_2 + U_3 + \dots + U_m\right) \cdot V^+$$

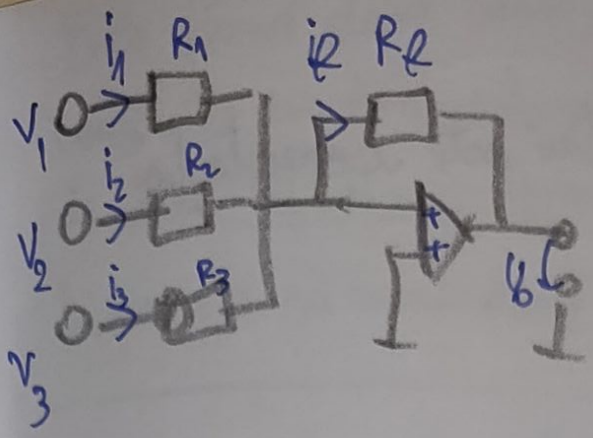
$$\frac{1}{R_p} = \frac{1}{R_1} + \frac{1}{R_2} + \dots + \frac{1}{R_m}$$

A.O. Sumator inversor (-)



$$U_0 = -R_f \sum_{i=1}^{n} \frac{U_i}{R_i}$$

daca $R_1 = R_2 = \ldots = R_n = R_f \Rightarrow U_0 = -\sum_{i=1}^{n} (-U_i)$



$$i_f = i_1 + i_2 + i_3$$

$$i_f = -\left(\frac{V_1}{R_1} + \frac{V_2}{R_2} + \frac{V_3}{R_3}\right)$$

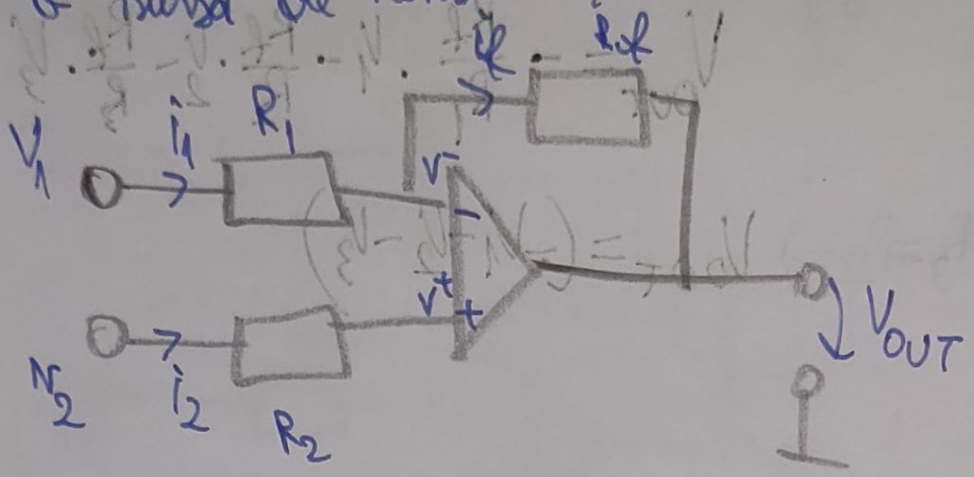
$$V_{OUT} = -\frac{R_f}{R_1} \cdot V_1 - \frac{R_f}{R_2} \cdot V_2 - \frac{R_f}{R_3} \cdot V_3$$

dacă $R_1 = R_2 = R_3 = R_f \Rightarrow V_{OUT} = (-V_1 - V_2 - V_3)$

$$V_{OUT} = -\left(V_1 \cdot \frac{R_f}{R_1} + V_2 \cdot \frac{R_f}{R_2}\right)$$

A.O. Diferențial

La fiecare bornă a A.O.-ului este conectată o sursă de tensiune/curent.



$$i_1 = \frac{V_1 - V^-}{R_1}, \quad i_2 = \frac{V_2 - V^+}{R_2}, \quad i_f = \frac{V^- - V_{OUT}}{R_f}$$

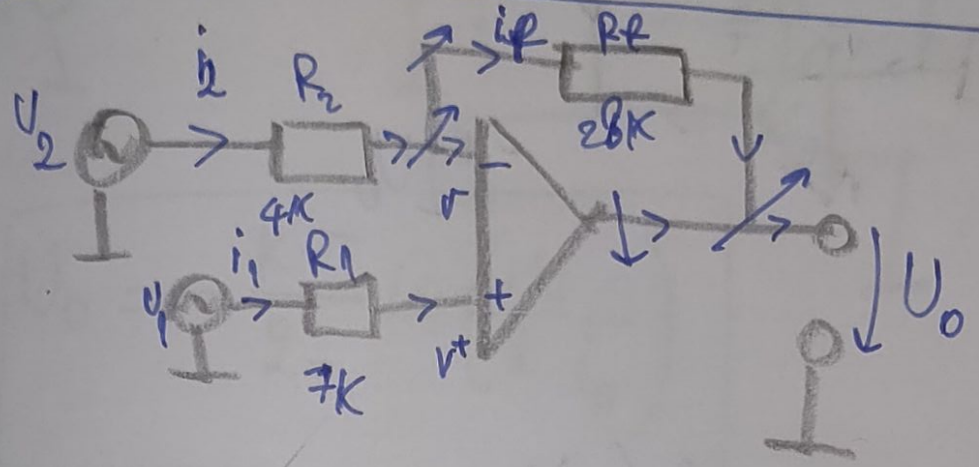
A.O. este în R.N. (Reacție Negativă) $\Rightarrow$ $\boxed{V^- = V^+}$

~~$V^+ = V_2$~~ $\Rightarrow$

$$\boxed{V_{OUT} = -\frac{R_f}{R_1} \cdot V_1 + \frac{R_f}{R_2} \cdot V_2}$$

Date fiind două surse de semnal $U_1(t)$ și $U_2(t)$, construiți un ~~A. O.~~ circuit cu A. O. care să genereze o tensiune $U_0(t) = 4 \cdot U_1(t) - 7 \cdot U_2(t)$.

Încercați să minimizați numărul de componente



A. O.-ul construit este în Reacție Negativă (R. N.) $\Rightarrow$

$\Rightarrow \boxed{V^- = V^+}$

Alegem $R_f = 28K\Omega$, $R_1 = 7K\Omega$, $R_2 = 4K\Omega$

$$U_0(t) = -\frac{R_f}{R_2} \cdot U_2(t) + \frac{R_f}{R_1} \cdot U_1(t)$$

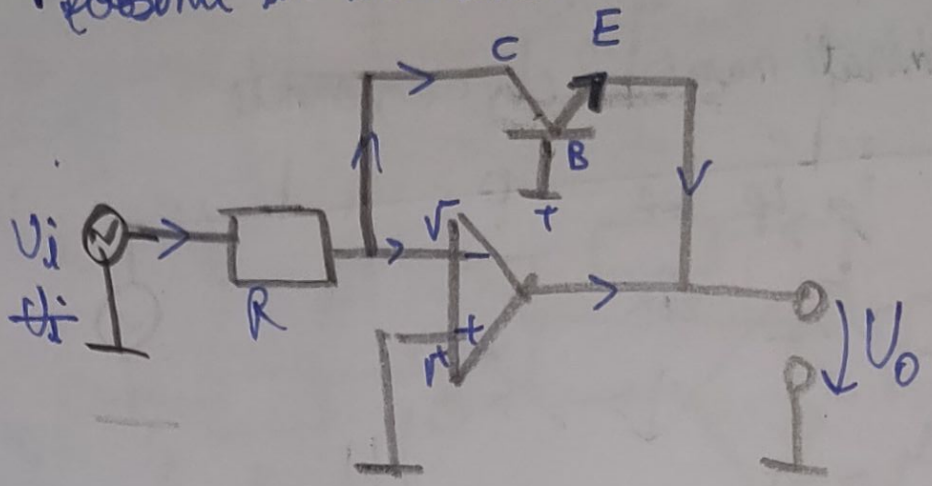
$$U_0(t) = -\frac{28K\Omega}{4K\Omega} \cdot U_2(t) + \frac{28K\Omega}{7K\Omega} \cdot U_1(t)$$

$$U_0(t) = -7U_2(t) + 4 \cdot U_1(t)$$

$$U_0(t) = 4 \cdot U_1(t) + 7 \cdot U_2(t)$$

Amplificatorul operațional logaritmic

- folosind un tranzistor

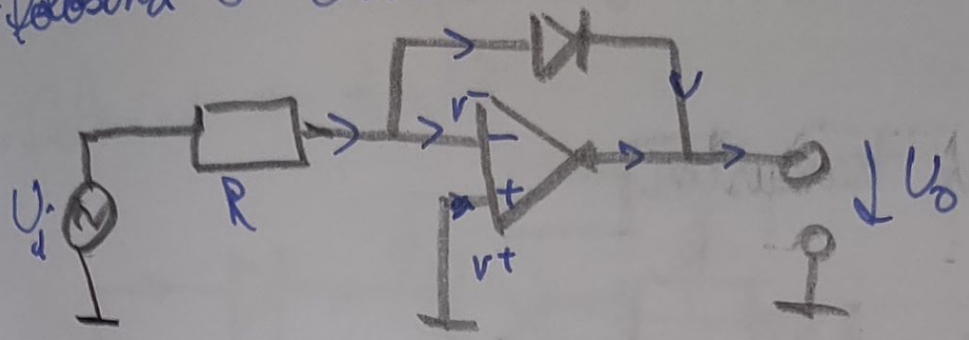


$$U_0 = -U_{BE} = -U_{th} \ln\left(\frac{i}{i_S}\right)$$

$$\boxed{U_0 = -U_{th} \cdot \ln\left(\frac{U_S}{R \cdot i_S}\right)}$$

Amplificatorul Operational Logaritmic

- folosind o diodă

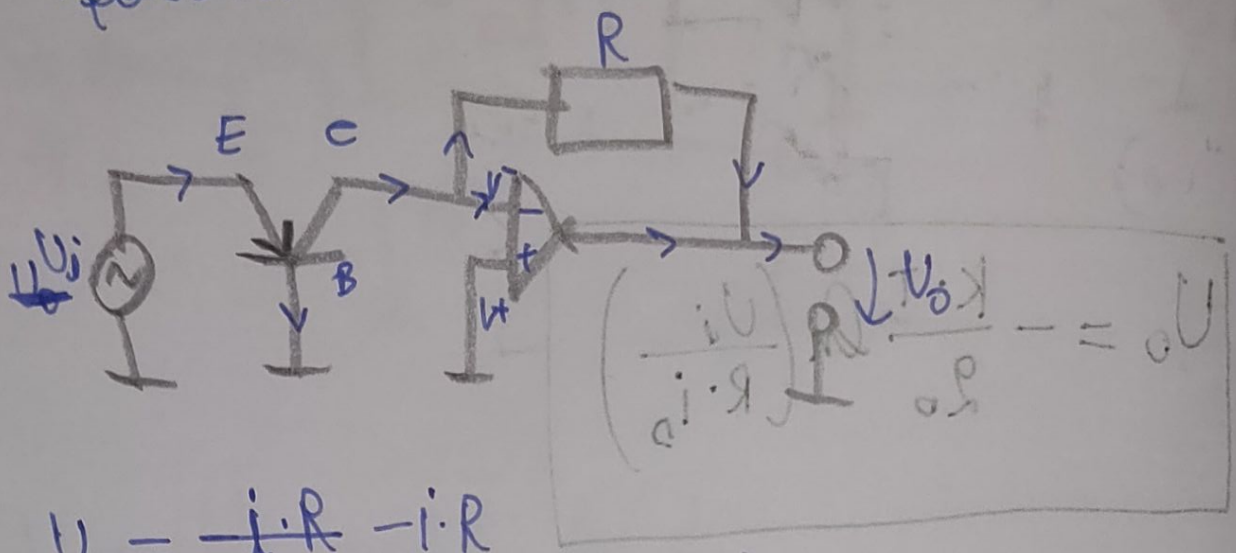


$$U_0 = -\frac{K \cdot T}{q_0} \ln\left(\frac{U_i}{R \cdot i_D}\right)$$

Amplificatorul Operațional Exponențial (A.O. Exp.)

(anti-logaritmic)

- folosind un tranzistor

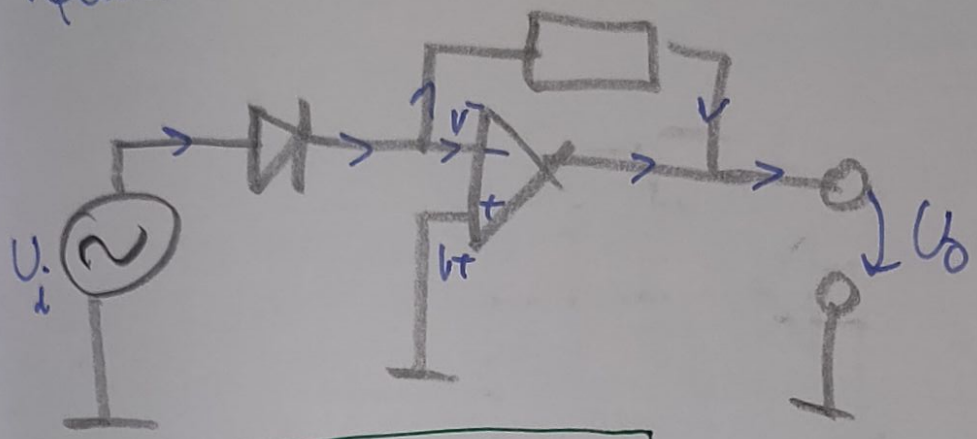


$$U_{ie} = \cancel{i \cdot R} - i \cdot R$$

$$i = i_S \cdot e^{\frac{V_{EB}}{V_{th}}} = i_S \cdot e^{\frac{V_S}{V_{th}}}$$

Amplificatorul Operațional Exponențial (A.O. Exp.)
(anti-logaritmic)

- folosind o diodă

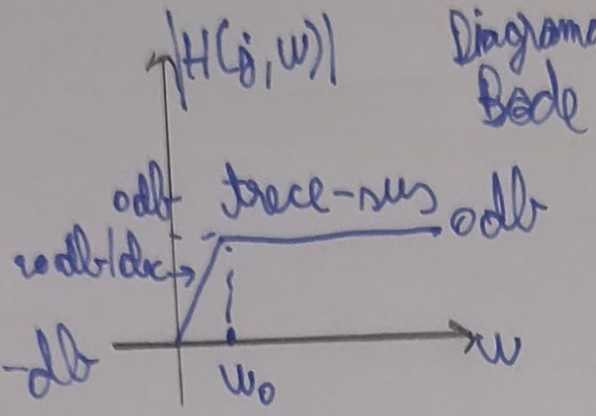


$$U_0 = -R \cdot i_S^{\frac{q_0 \cdot U_i}{K \cdot T}}$$

Filtre

Filtru trece-sus (F.T.S.)

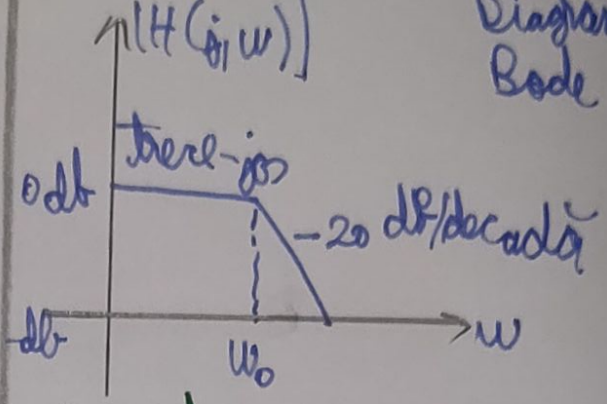
Diagrama Bode



$$\omega_0 = \frac{1}{R \cdot C}$$

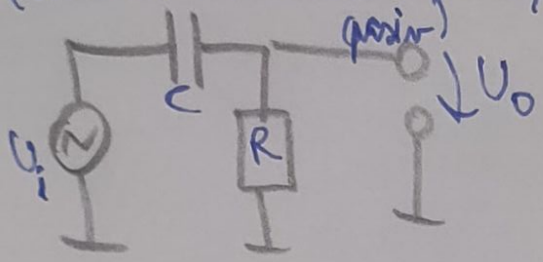
Filtru trece-jos (F.T.J.)

Diagrama Bode



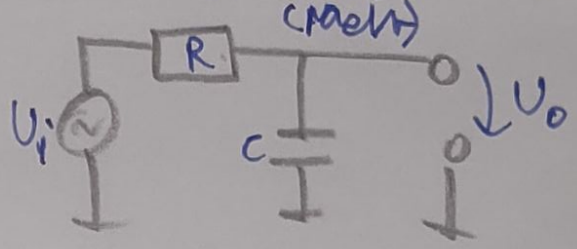
$$\omega_0 = \frac{1}{RC}$$

Filtru trece-sus (fără impedanță)



$$\omega_0 = \frac{1}{R \cdot C}$$

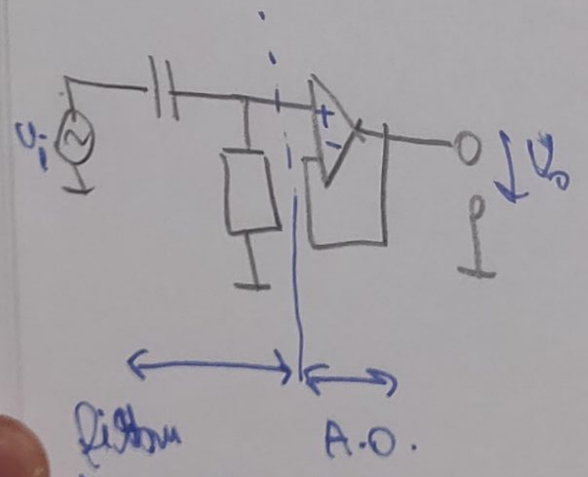
Filtru trece-jos (fără impedanță)



$$\omega_0 = \frac{1}{R \cdot C}$$

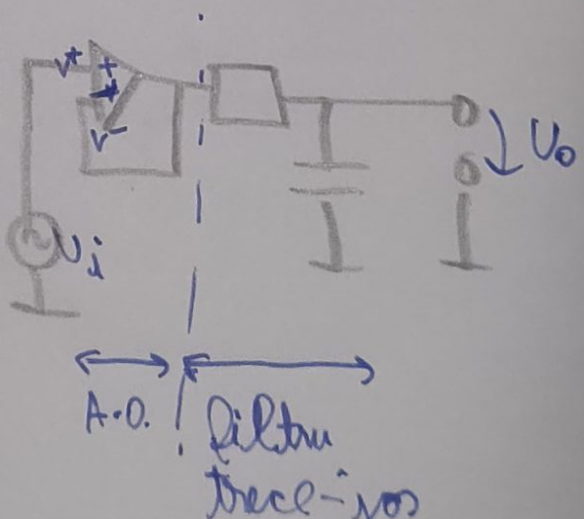
Filtru trece-jos (cu impedanță) (activ)

• adăugăm A.O. (repetor)



Filtru trece-jos (cu impedanță) (activ)

• adăugăm A.O. repetor tensiune

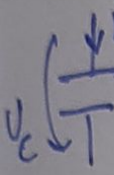


impedanta = un fel de rezistenta

$\frac{1}{T}$ $\frac{1}{R \omega C}$ $\xrightarrow{\text{Laplace}}$ $\frac{1}{C \cdot D}$ 

R $\xrightarrow{\text{Laplace}}$ R 

$R \cdot \omega C \xrightarrow{\text{Laplace}} L \cdot D$ 



condensator $\boxed{i_c = C \frac{dV_c}{dt}}$ → trece curent alt. ~

→ Nu trece curent $\pm T$ cont. $-T$

$i_c > 0 \Leftrightarrow V_c$ crește ↑

$i_c < 0 \Leftrightarrow V_c$ scade ↓

$i_c = 0 \Leftrightarrow V_c$ rămâne constant

$[C]_{SI} = F$ (Farad)

Bobina (inductoare)

~ — sursă de curenta alternativ

$i(t) = A \cdot \sin(\omega \cdot t + \varphi_0)$

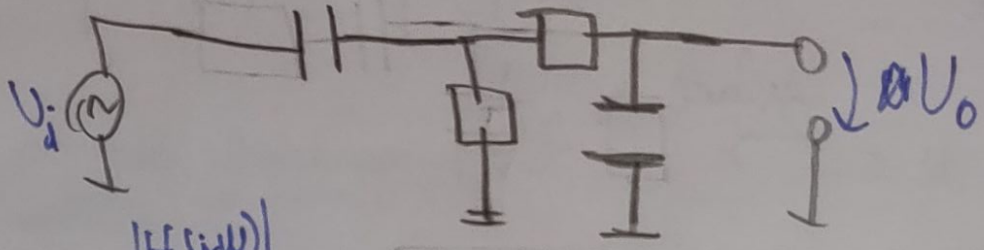
$A = I_{max}$ = amplitudinea

ω = frecventa anguläră, pulsația

$\omega = 2 \cdot \pi \cdot f$, f = frecventa (de frângere), $f = \frac{1}{T}$

φ_0 = faza inițială, unghi $\in [0, 2\pi]$

Filtru de bandă pasiv = trece sus + trece jos



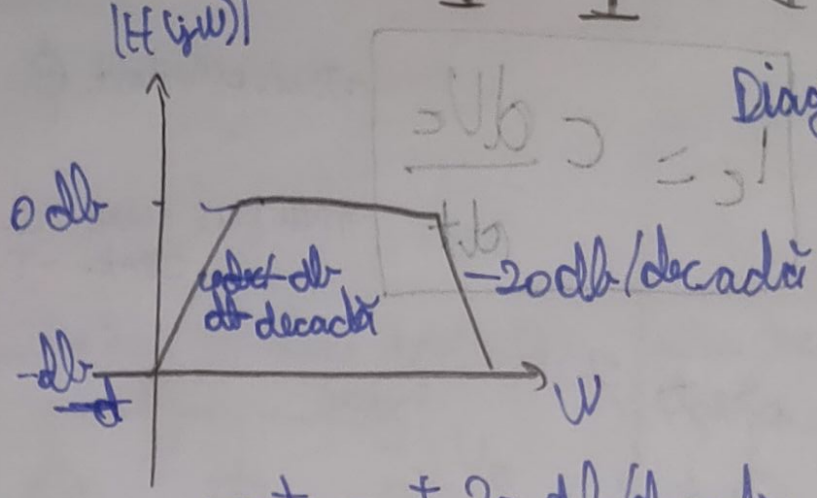
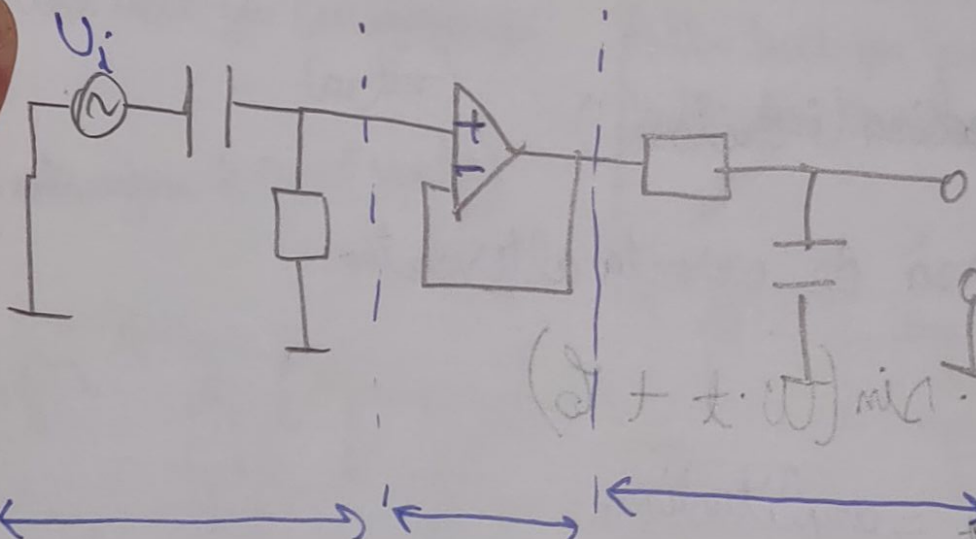


Diagrama Bode

panta = ± 20 db/decadă

Filtru de bandă activ

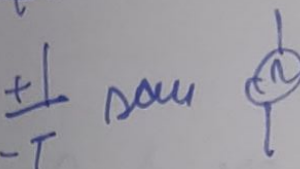


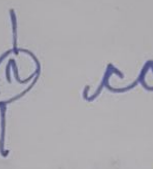
filtru trece-sus (F.T.S.)

A.O. (repetor) sau neinversor repetor

filtru trece-jos (F.T.J.)

A.O. reduce rezistența internă a surselor de tensiune
De ce avem nevoie de A.O. (impedanță infinită)?
. În cazul în care apar modificări de la alte componente externe, tensiunea circuitului nostru să rămână constantă
. fără un A.O. repetor, ar intra rezistența filtrului în serie / paralel cu a generatorului



sau  care nu este ideal,
~~alte~~ surselor au și ele o rezistență internă
și fără A.O., D-ar modifica frecvența de frângere

Tipuri de filtre:

- trece jos pasiv
- trece sus pasiv
- bandă pasiv
- trece jos activ
- trece sus activ
- bandă activ

Un filtru pasiv nu are impedanță, nu are A.O.
(A.O. = Amplificator Operațional)

Un filtru activ are impedanță, are A.O.
(A.O. = Amplificator Operațional)

Pentru a obține un filtru activ, este suficient să adăugăm în circuit un A.O. repetor de tensiune.

filtru de bandă = filtru trece-jos + filtru trece-sus

04

Construiți un filtru trece-jos cu frecvența de tăiere $f_0 = 1000$ Hz și alegeți valorile componentelor necesare.

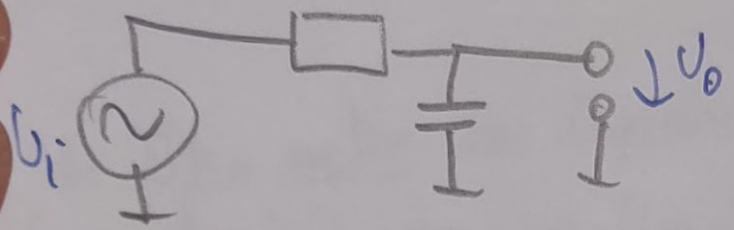
Îmbunătățiți filtrul pentru a avea impedanță de intrare infinită.

Dați un exemplu unde acest lucru este important?

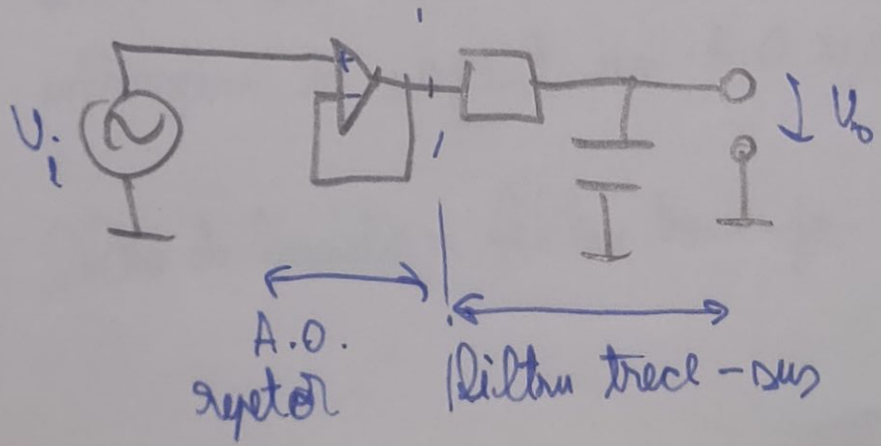
$\omega_0 = 2\pi f_0 = 2\pi \cdot 1000 = 6,28 \cdot 10^3$

$U_i(t) = I_{max} \cdot \sin(\omega_0 t + \varphi_0)$

Filtru trece-jos fără impedanță



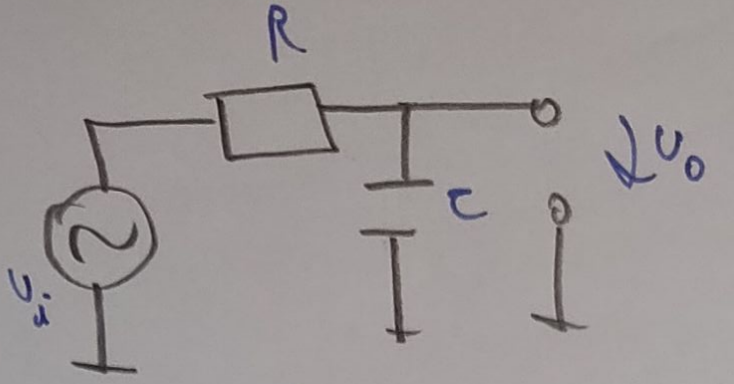
Filtru trece-sus cu impedanță

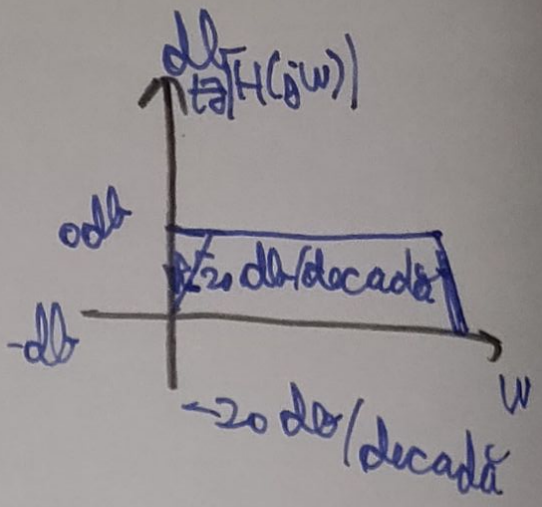


Exemple de situații în care un filtru trece-jos cu impedanță infinită:

- filtrare de zgomot în senzori → unde se dorește eliminarea semnalelor cu frecvențe mari
- filtre împotriva semnalelor de înaltă frecvență
- filtrare în sisteme audio
- instrumente medicale, ex.: ECG

Filtros trace-jos (F. T. J.)

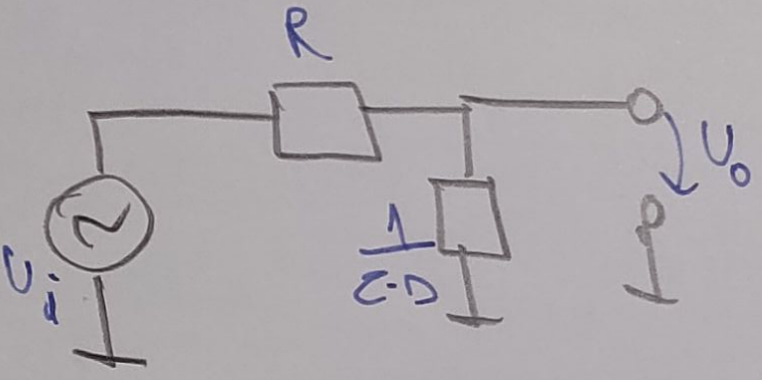




$R \rightarrow R$

$C \rightarrow \frac{1}{C \cdot D}$

$L \rightarrow L \cdot D$



$$U_o(D) = U_i(D) \cdot H(D) = U_i(D) \cdot \frac{1}{1 + R \cdot C \cdot D}$$

$$H(D) = \frac{U_o(D)}{U_i(D)} = \frac{1}{1 + R \cdot C \cdot S}$$

$R = 1k\Omega$

$C = 1\mu F$, $\omega = 10^5$

$v_i(t) = \sin(\underbrace{10^5}_{\omega} t)$

$$v_0(t) = |H(j\omega)| \cdot \sin(\omega t + \angle H(j\omega))$$

$$A = \left| \frac{1}{1 + R \cdot C \cdot j\omega} \right|$$

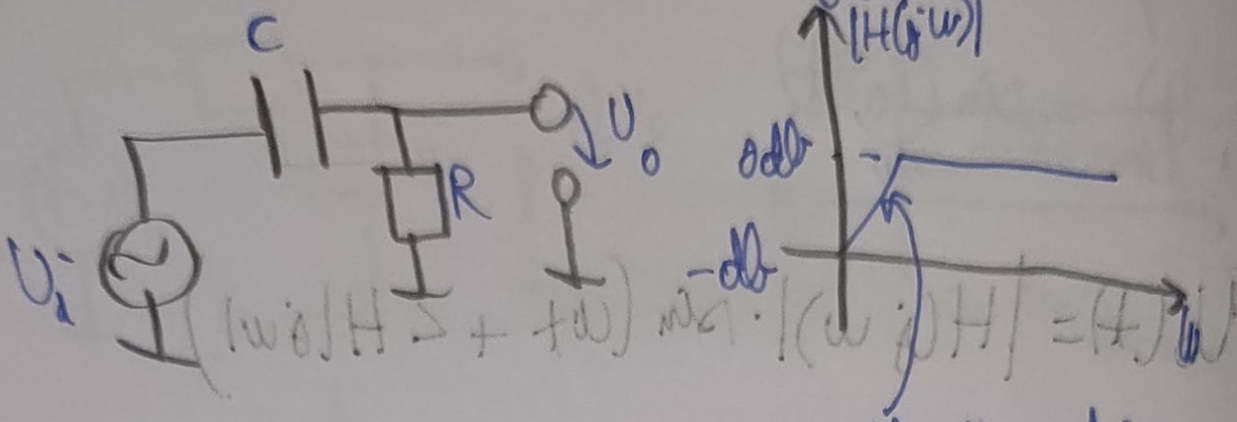
$A = |H(j\omega)|$ amplitudinea

$$A = \left| \frac{1}{1 + 10^3 \cdot 10^{-5} \cdot 10^5 j} \right| = \left| \frac{1}{1 + 100j} \right| \simeq \frac{1}{100} =$$

$$(a)H \cdot (a)_i U = (a)_o U$$

$$\frac{a \cdot \Omega \cdot 9}{a \cdot \Omega \cdot 9 + 1} = \frac{(a)_o U}{(a)_i U} = (a)H$$

Filtro trece-basso (F.T.S.)

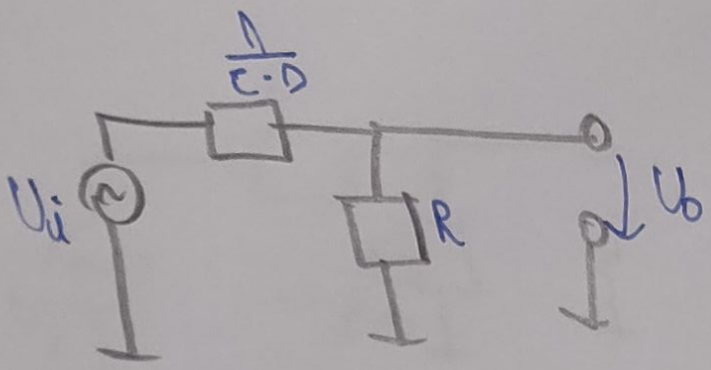


$$\left|\frac{1}{1+j\omega RC}\right| = A$$

$$|H(j\omega)| = A$$

$C \rightarrow \frac{1}{T} \Rightarrow \frac{1}{C \cdot D}$

$L \rightarrow \Rightarrow L \cdot D$



$$U_o(D) = U_i(D) \cdot H(D)$$

$$H(D) = \frac{U_o(D)}{U_i(D)} = \frac{R \cdot C \cdot D}{1 + R \cdot C \cdot D}$$

$$v_0(t) = |H(j\omega)| \cdot \sin(\omega t + \angle H(j\omega))$$

$R = 1K\Omega$

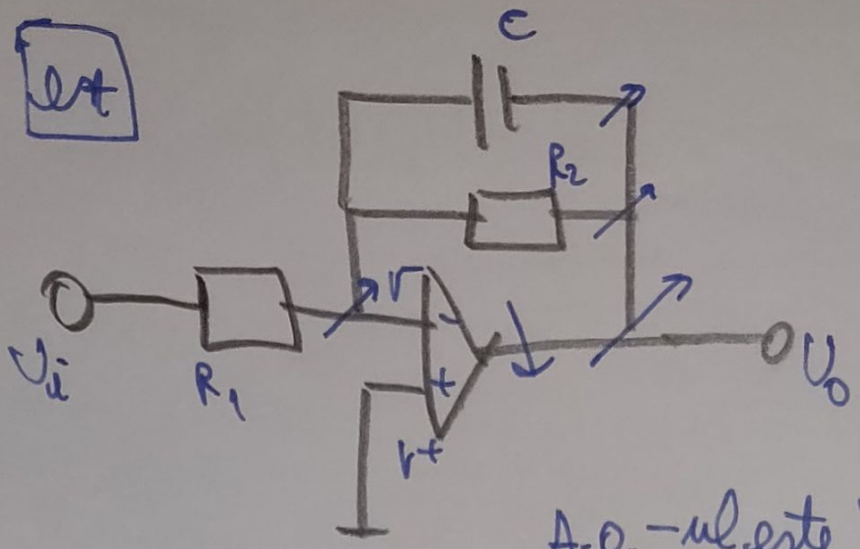
$C = 1\mu F$

$v_i(t) = \sin(\underbrace{10^5}_{\omega} t)$

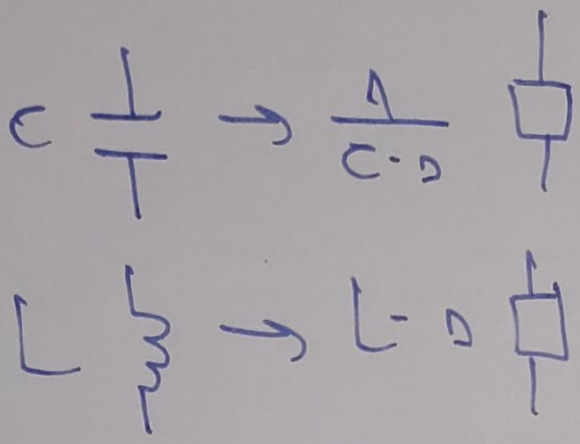
$$A = |H(j\omega)| = |H(j\cdot 10^5)| = \left| \frac{10^3 \cdot 10^{-5} \cdot 10^5 j}{1 + 10^3 \cdot 10^{-5} \cdot 10^5 j} \right| =$$

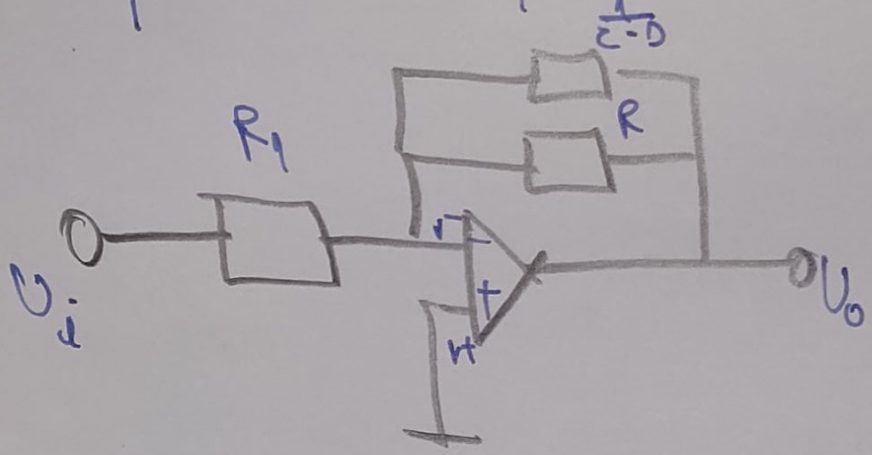
$$= \left| \frac{100j}{1 + 100j} \right| \simeq 1$$

a)



A.O. – ul este în R.N. Reacție negativă





$$R_e = \frac{1}{sC} \| R_2$$

$$\frac{1}{R_e} = \frac{1}{\frac{1}{sC}} + \frac{1}{R_2} \Rightarrow \frac{1}{R_e} = sC + \frac{1}{R_2} \Rightarrow \frac{1}{R_e} = \frac{R_2 sC + 1}{R_2}$$

$$R_e = \frac{R_2}{R_2 sC + 1}$$

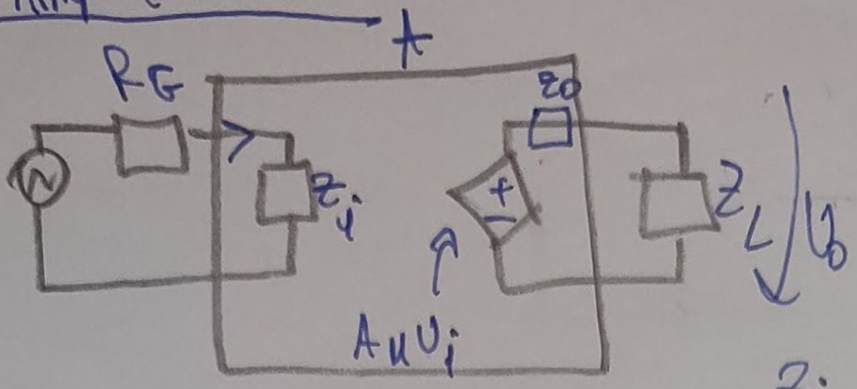
$$R_e = \frac{R_2}{1+R_2 \cdot S \cdot C}$$

$+$ A.O. in R.N. $\Rightarrow V^+ = V^-$

$$\frac{V_i - 0}{R_1} + \frac{V_o - 0}{\frac{R_2}{1+RSC}} = 0$$

$$V_o = -\frac{V_i}{R_1} \cdot \frac{R_2}{1+R_2 SC} \Rightarrow H(S) = \frac{V_o}{V_i} = -\frac{R_2}{R_1(1+R_2 SC)}$$

Amplificatoare



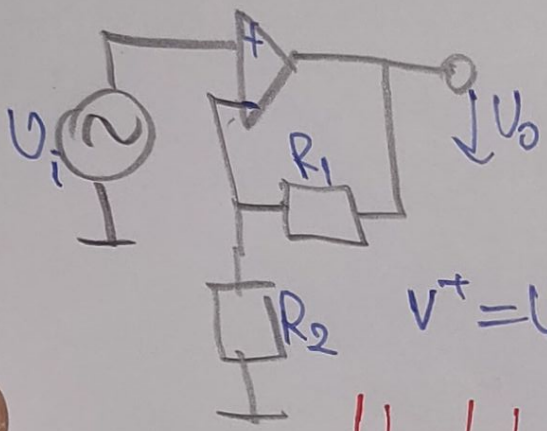
$$U_o = U_i \cdot A_u \cdot \frac{z_i}{R_G + z_i} \cdot \frac{z_L}{z_L + z_{ie}}$$

$$A_u = \frac{U_{o\infty}}{U_2}$$

$$z_i = \frac{U_i}{i_i}, \quad z_o = \frac{A_u U_o}{I_{D_{sc}}}$$

$$z_o = \frac{U_{o\infty}}{i_{sc}}$$

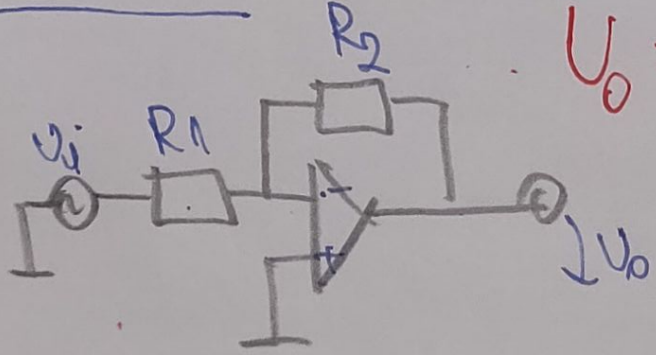
Neinversor



$$V^+ = U_i$$

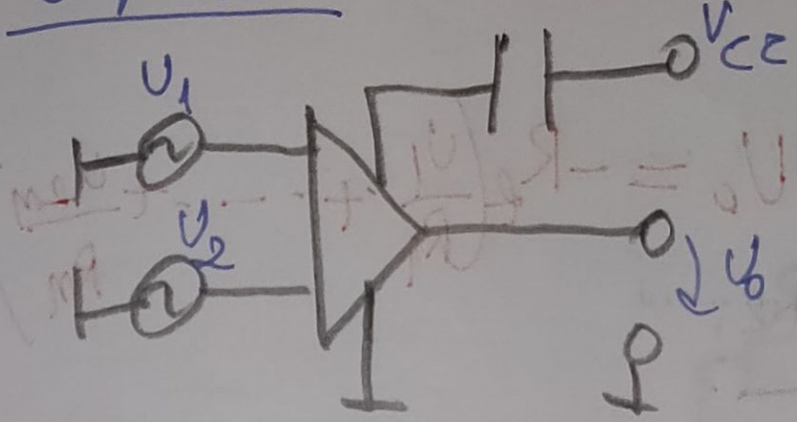
$$U_o = U_i \cdot \frac{R_1 + R_2}{R_2}$$

inversor



$$U_o = -U_i \cdot \frac{R_2}{R_1}$$

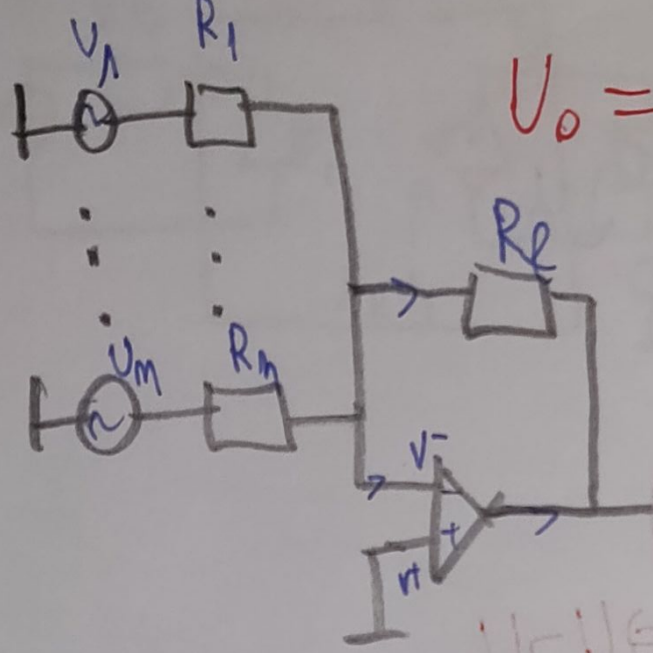
Comparator



$$U_0 = A \cdot (U_1 - U_2)$$

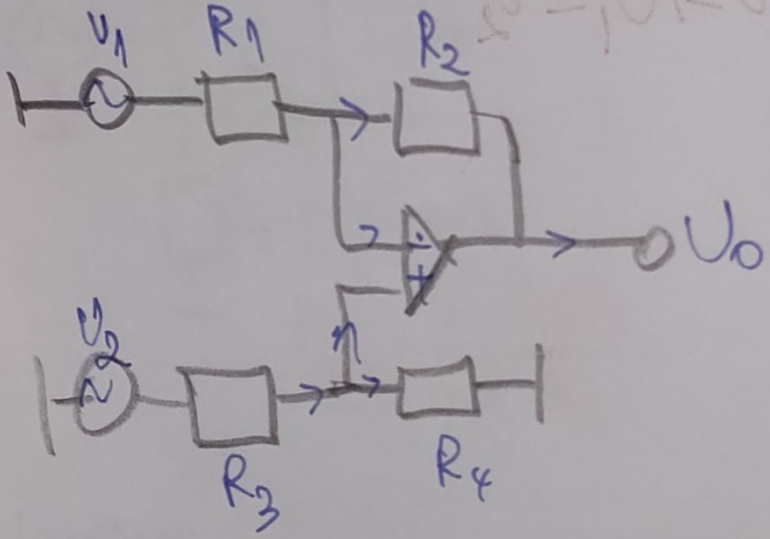
- $A > 0 \Rightarrow U_1 > U_2$
- $A < 0 \Rightarrow U_2 > U_1$
- $A = 0 \Rightarrow U_1 = U_2$

Sumator



$$U_0 = -R_\ell \left(\frac{U_1}{R_1} + \cdots + \frac{U_m}{R_m} \right)$$

Diferențial

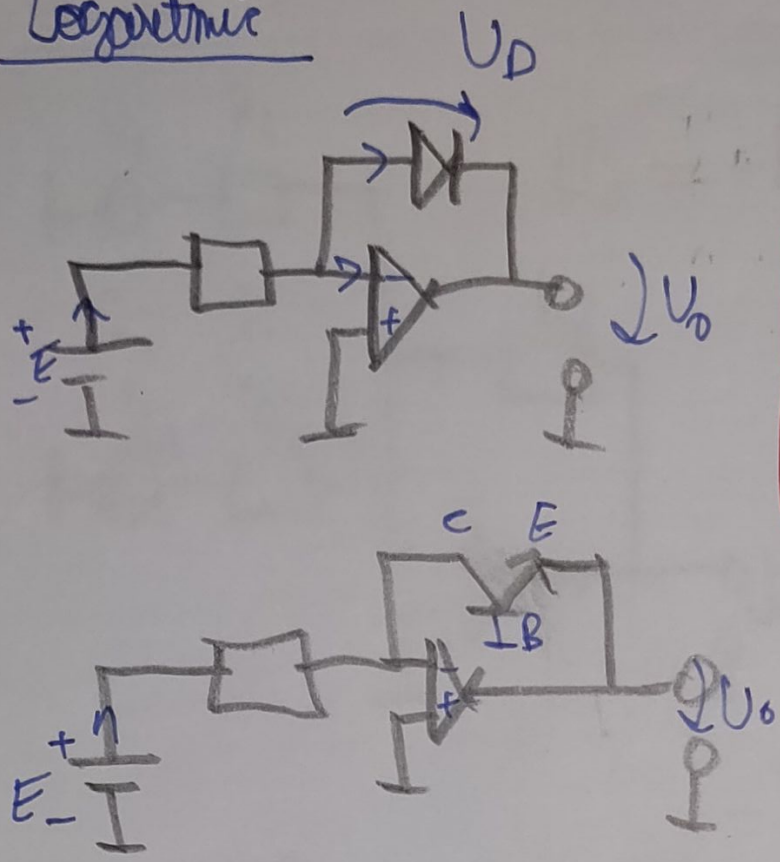


$$U_0 = \left(1 + \frac{R_2}{R_1}\right)\left(\frac{R_4}{R_3 + R_4}\right) U_2 - \frac{R_2}{R_1} \cdot U_1$$

Pentru $R_1 = R_3$ și $R_2 = R_4 \Rightarrow$

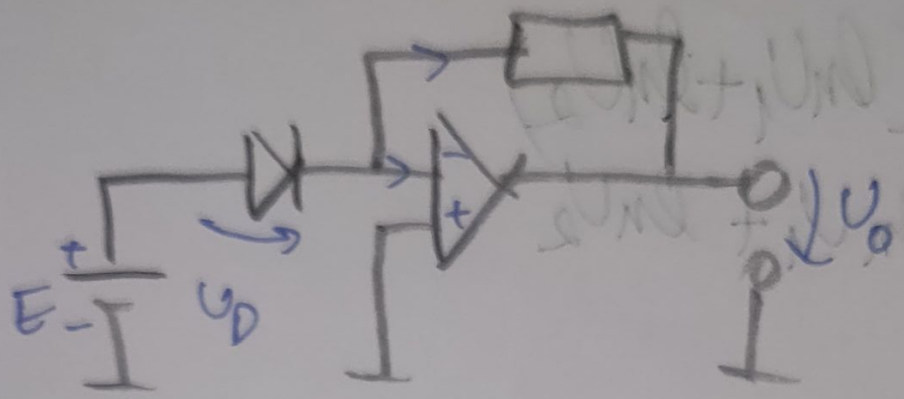
$$U_0 = \frac{R_2}{R_1} (U_2 - U_1)$$

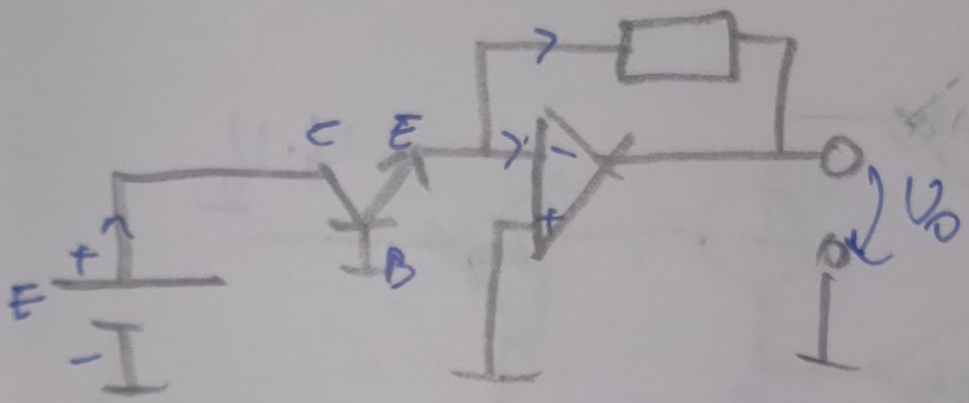
Logaritmic



$$U_0 = -\ln \frac{U_i}{R}$$

Exponential

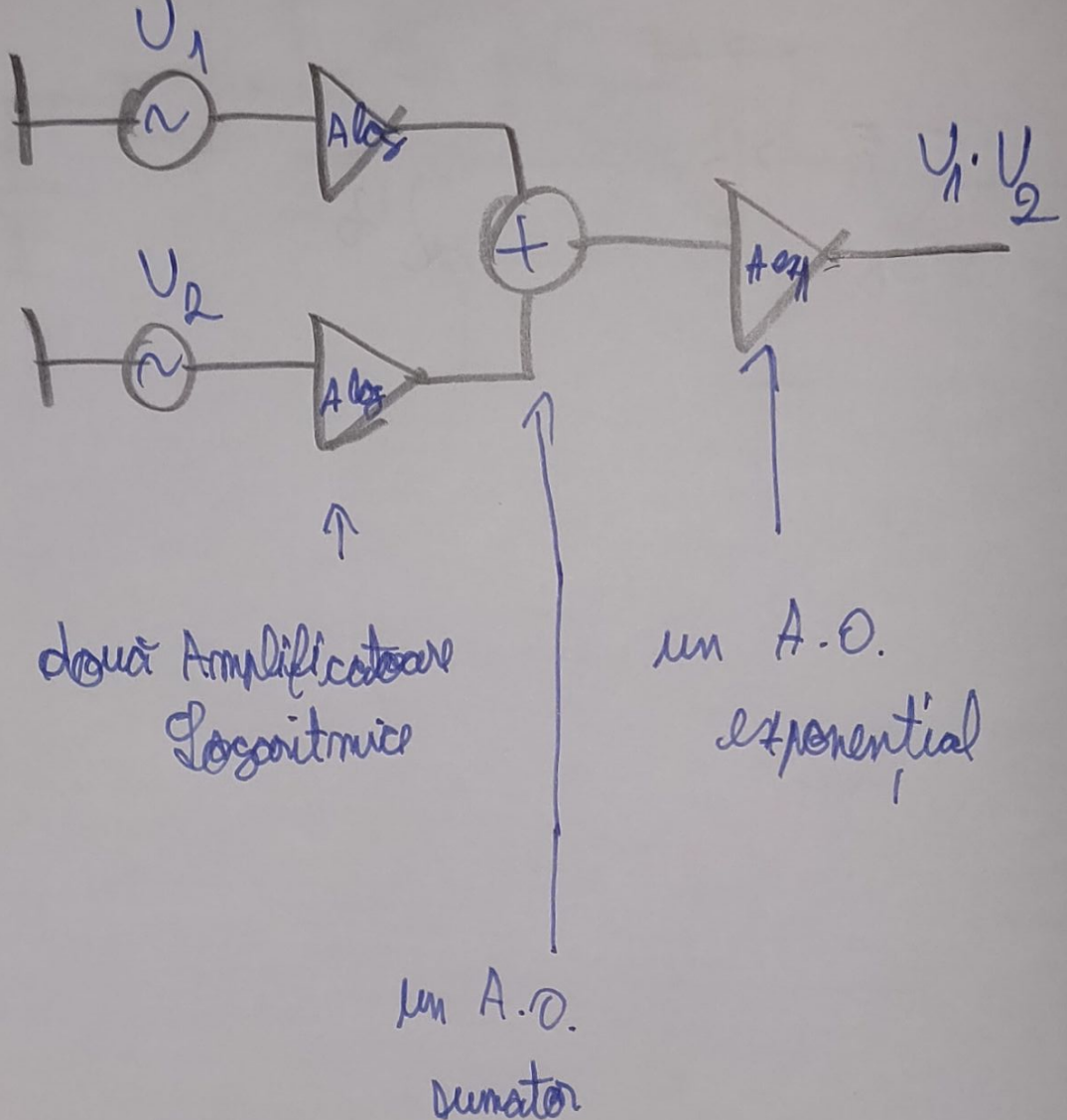


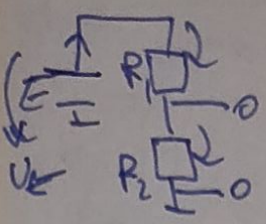


A.O. înmulțire U_1 și U_2

$$U_1 \cdot U_2 = \exp(\ln U_1 + \ln U_2)$$

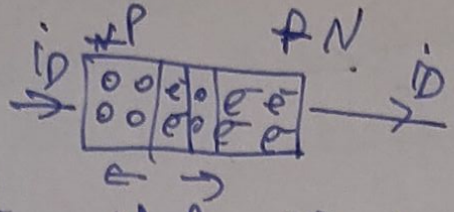
$$U_1 \cdot U_2 = e^{\ln U_1 + \ln U_2}$$



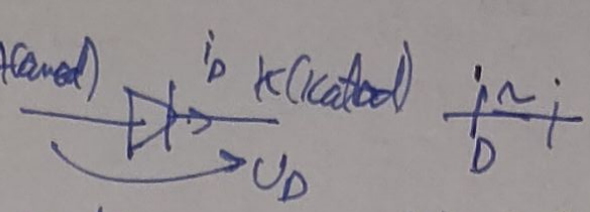


divizor de tensiune

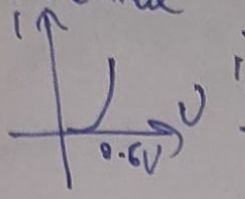
$$\frac{E_e}{R_e} = \frac{E}{R_1} \quad ; \quad \frac{1}{R_e} = \frac{1}{R_1} + \frac{1}{R_2}$$



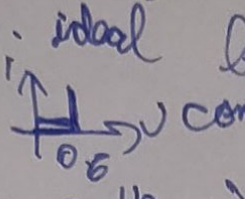
diodă: i_D → P | N → i_D



A(anod) i_D K(catod) U_D $i \sim i_D$



normal: i_D vs U, 0.6V

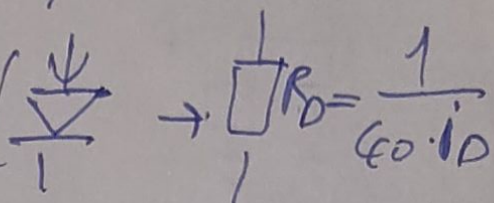


ideal: i_D vs U, 0.6

blocată: $i_D = 0$, $U_D = 0.6V$

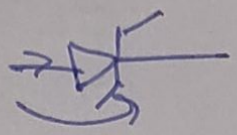
conducție directă: $i_D > 0$, $U_D = 0.6V$

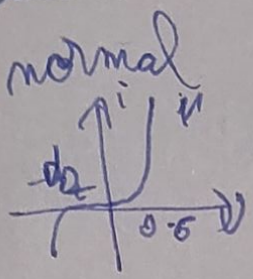
$$i_D = i_S\left(e^{\frac{q_0 U_D}{kT}} - 1\right) \simeq i_S e^{\frac{1}{25mV} \cdot U_D}$$



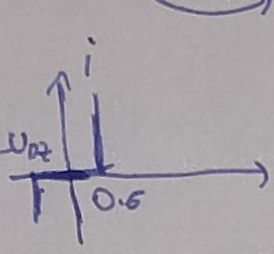
$$R_D = \frac{1}{q_0 \cdot i_D}$$

diodă Zener





normal: i vs U, U_{DZ}, 0.6



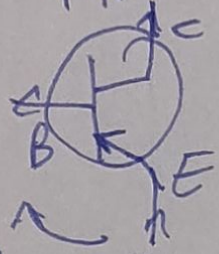
i vs U, U_{DZ}, 0.6

blocată: $-U_{DZ} < U_D < 0.6V$, $i_D = 0$

cond. dir.: $U_D = 0.6V$, $i_D > 0$

cond. inv.: $U_{DZ} < U_D$ $U_D = -U_{DZ}$ i_D ←

PNP



C, B, E

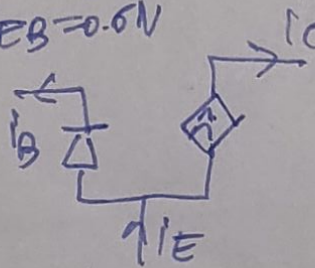
C → P → gol

B → N → e^-

C → P → gol

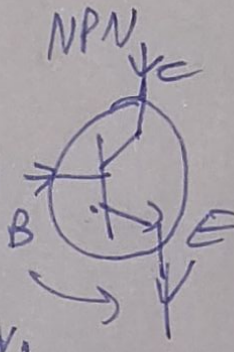
RAN: $V_{EC} > V_{EB} = 0.6V$

$V_{EB} = 0.6V$



i_C, i_B, i_E

NPN



C, B, E

E → N → e^-

B → P → gol

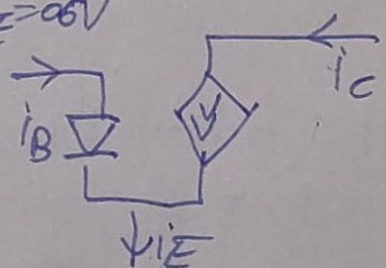
E → N → e^-

$i_E = i_B + i_C$

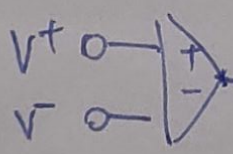
$i_C = \beta \cdot i_B$

RAN: $U_{CE} > U_{BE} = 0.6V$

$U_{BE} = 0.6V$



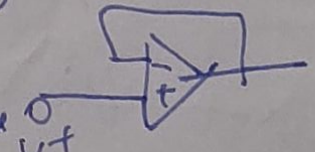
i_C, i_B, i_E



V^+, V^-, V_0

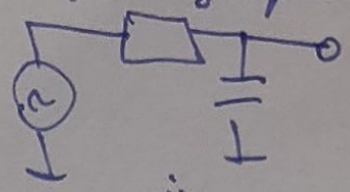
$U_0 = A(V^+ - V^-)$

R.N. ⇐⇒ ieșirea e întoarsă la −

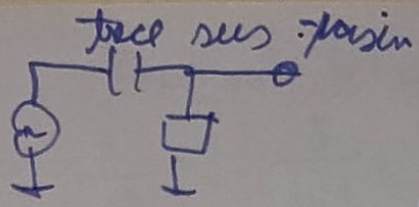


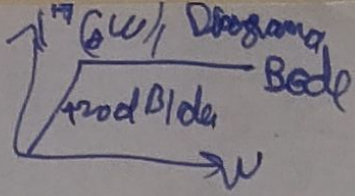
V^+

trece jos pasiv

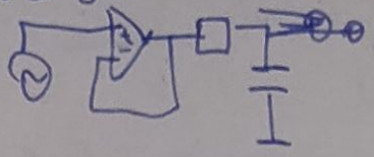


trece sus pasiv

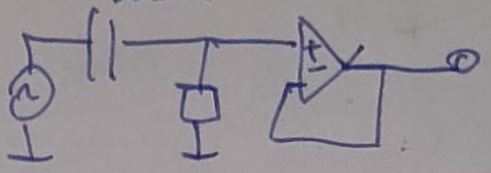


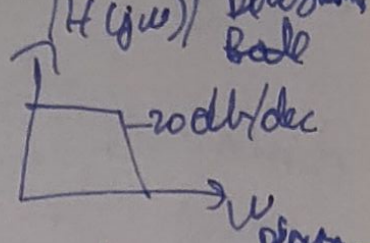


trece jos activ

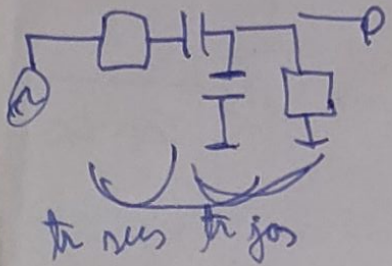


trece sus activ

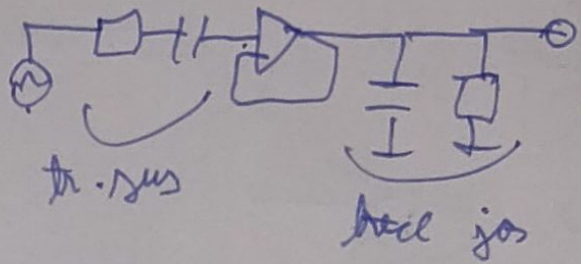


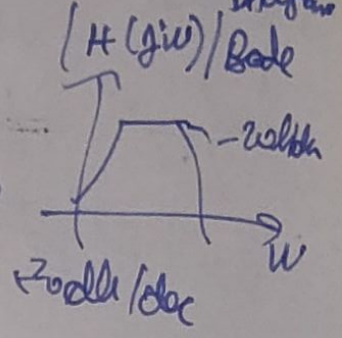


Banda pasiv



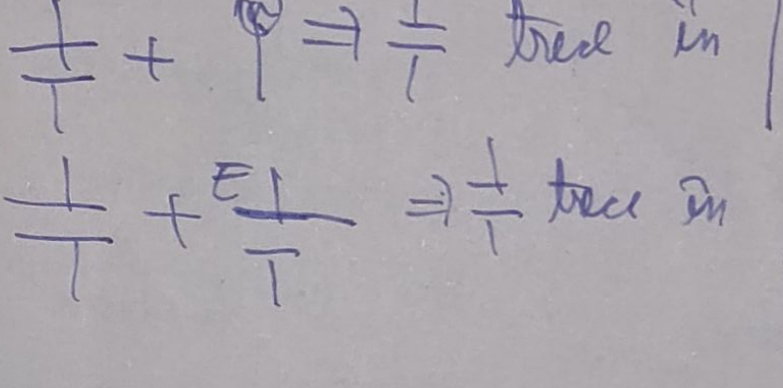
Banda activ





liniarizare $U_0(U_1, U_2, E) = U_0(0,0,E) + U_0(0, U_2, 0) + U_0(U_1, 0, 0)$

pasivizam:



$H(s) = \frac{U(s)}{I(s)}$

$R \to R$

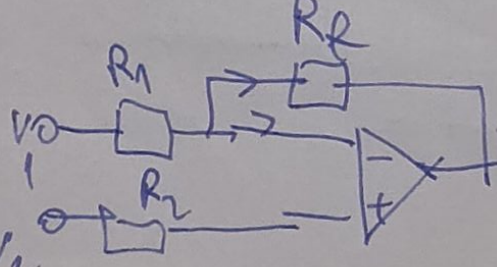
$L \to sL$

$C \to \frac{1}{sC}$

$u(t) = A \sin(\omega t + \varphi_0)$

$\omega = 2 \cdot \pi \cdot f$; ω = pulsatia

f = frecventa



$V_0 = -\frac{R_f}{R_1} V_1 + \frac{R_f}{R_2 R_2} \cdot V_2$